

Received 13 August 2025, accepted 5 September 2025,
date of publication 10 September 2025, date of current version 17 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3608623

 SURVEY

A Comprehensive Survey on Pre Versus Post Quantum Security Schemes for 5G-Enabled IoT Applications

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This work was supported in part by the Central Institute of Technology Kokrajhar, India, through PDA Grant.

ABSTRACT The proliferation of 5th generation (5G) technology has catalyzed significant advancements in various application of Internet of Things (IoT) such as Intelligent Transportation Systems, Smart Energy, Industrial IoT (IIoT), Digital Twins, Autonomous Aerial Vehicles (UAVs) and Smart Health. However, with the upcoming advent of quantum computing, conventional security methods are encountering new and significant challenges, requiring the revaluation of pre and post-quantum security measures. This study provides an in-depth analysis of security measures before and after the advent of quantum computing for intelligent IoT applications enabled by 5G-connected network. The survey thoroughly analyzes the strengths and weaknesses of conventional public key cryptography methods such as RSA, Diffie-Hellman, Secret Sharing, and Elliptic Curve Cryptography (ECC), which are susceptible to quantum risks, and postquantum cryptographic methods such as lattice-based cryptography, code-based cryptography, and multivariate based techniques, which may provide protection against quantum attacks. Furthermore, we evaluate the suitability of these strategies in the context of 5G-enabled IoT deployments, taking into account the distinct requirements and obstacles presented by interconnected IoT devices. Although pre quantum public key cryptographic algorithms are still in use, but active research is going on to design and implement PQC algorithms to secure 5G IoT, while hybrid schemes have shown great promise being backward compatible with traditional cryptographic approaches at the same time bring quantum resistance. The survey offers significant insights into the latest pre and post-quantum security solutions for 5G-enabled IoT Applications. It helps in making educated decisions and implementing strong security measures in preparation for the quantum computing age.

INDEX TERMS Internet of Things, 5G, digital twin, ECC, RSA, post-quantum, lattice based cryptography, code based cryptography, multivariate cryptography.

I. INTRODUCTION

Recently, the Internet of Things(IoT) has completely changed every aspect of human life with advance communication technologies. IoT connects a large number of objects or things such as Radio-Frequency IDentification (RFID)

The associate editor coordinating the review of this manuscript and approving it for publication was Mohamad Afendee Mohamed^{1b}.

tags, sensors, actuators, smart devices, etc. – which, are able to interact with each other to achieve varied goals that improve the standard of human living [1]. It has been estimated that by the end of 2025, 75.44 billion devices would be connected [2]. As a consequence, IoT is progressively becoming recognized as the next stage in the evolutionary growth of the Internet. The ushered growth of 5th generation (5G) has revolutionized several domains

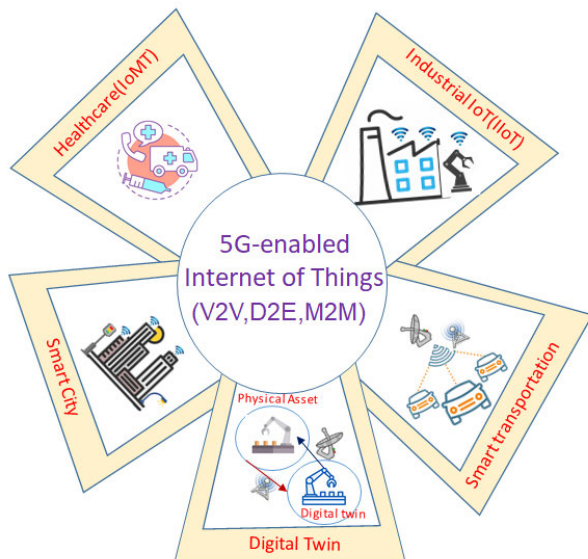


FIGURE 1. 5G-enabled IoT Applications.

of digital communication systems. The integration of 5G technology into Internet of Things (IoT) ecosystems marks a significant advancement in smart applications. 5G network offers very high speed with low latency, which are critical for the vast network of IoT devices that depend on flawless communication to operate [3]. The enhanced capabilities of 5G is accelerating the growth of IoT applications in [4], [5], [6], and [7] several fields such as healthcare (Internet of Medical Things (IoMT) [8]), intelligent vehicle monitoring (Internet of Vehicular Things (IoVT) [9]), submarines, and underwater explorations (Internet of Underwater Things (IoUT) [10]), smart home [11], grid [12], supply chain [13], and precision agriculture [14]. Figure 1 shows the Vehicle to Vehicle (V2V), Device to Everything (D2E) and Machine to Machine (M2M) communication in 5G-enabled IoT. The biggest significance of this adoption is that the intervention of humans is greatly reduced, thereby resulting in a sharp decline in manual errors. It is quite evident that applications of IoT are not limited to industrial and commercial sectors; it has a vivid and varied presence in hospitals, where it is used as a tool to monitor patients and assist doctors, as well as in some of our daily possessions like watches, cars, and portable electronic devices that are battery-operated [5]. Since most of the devices are battery-operated, they have limited power supply [15]. As the devices are task-specific, they have limited computation and memory capabilities [3]. More often, IoT devices are called resource-constrained devices. As a result, these applications cannot accommodate complex algorithms for optimization [16]. With the exponential growth of 5G and IoT devices, the reliability of IoT services has increased. It is quite difficult to visualize a day without technology. This reliability has served as bait for many attackers and intruders who intend to exploit the network for their concealed motives of causing financial losses to an organization, individual, or company or to steal confidential

data as the devices are deployed at remote locations, i.e., out of sight of the user. Attackers may harm the devices physically, inject malicious software (Malware) to corrupt the system, or steal personal information, including identities, leading to breaches of privacy [17]. IoT security should, as a result, be a top priority. As IoT systems employ resource-constrained hardware, traditional security measures may not be compatible. Specialized resource-efficient security strategies and architectures are required for devices with limited resources. Another significant issue with the IoT network is the absence of backup security [17]. We must ensure a certain level of trust and security. In the recent past, many researchers have proposed novel strategies using public key cryptography tools to overcome certain issues in 5G-enabled IoT systems, with their primary aim being to tackle the issues of authentication, authorization, access controls, and preserving data confidentiality and privacy. However, in this age of quantum computing, there is an urgent need for thorough assessments of security measures developed to deal with the particular needs and obstacles of 5G-enabled IoT systems. The study will shed light on the effectiveness of security measures in reducing cyber risks in 5G-enabled IoT environments by analyzing the pros and cons of current cryptographic methods and post-quantum cryptography approaches. Moreover, we also discuss the potential solutions proposed in the recent past and analyze their pros and cons with respect to IoT security. For the ease of following, we have added the list of abbreviations used in this paper in Table 1.

A. MOTIVATION, SCOPE, AND CONTRIBUTIONS

Numerous benchmark surveys examining the susceptibilities of IoT systems to attacks can be found in the literature. However, current research does not provide thorough comparative assessments of security strategies before and after quantum computing, specifically designed for 5G-enabled IoT. Few studies [18], [19], [20], [21] directly evaluate the effectiveness, efficiency, and application of pre-quantum and post-quantum encryption in IoT environments, although examining them individually in the context of IoT or 5G. As a result, more studies comparing pre and post-quantum security approaches addressing the following research gap are necessary to better secure Internet of Things (IoT) devices, data transmission, and cloud-based services in 5G-enabled IoT systems:

- **Heterogeneity and Large Scale:** Finalization of pre- and post-quantum security schemes for protecting devices, data transmission, and cloud-based services in 5G-enabled IoT ecosystems with diverse devices, communication protocols, and data types.
- **Implementation Challenges:** Integrating post-quantum security measures in actual IoT installations, taking into account variables like computational complexity, interoperability, and backward compatibility is still an open area of research.

TABLE 1. List of abbreviations.

Symbol	Meaning
AES	Advanced Encryption Standard
AKC	Asymmetric Key Cryptographic
CoAP	Constrained Application Protocol
CVP	Closest Vector Problem
D2E	Device to Everything
DDoS	Distributed Denial of Service
DH	Diffie-Hellman
ECC	Elliptic Curve Cryptography
ECSSS	Elliptic Curve Secret Sharing Scheme
EDHOC	Ephemeral Diffie-Hellman over COSE
FDI	False Data Injection
FHE	Fully Homomorphic Encryption
GP	Gaussian pyramid
GPU	Graphical Processing Unit
HE	Homomorphic Encryption
HQC	Hamming Quasi Cyclic KEM
HTTP	Hyper Text Transfer Protocol
IBE	Identity-based Encryption
IIoT	Industrial Internet of Things
IoMT	Internet of Medical Things
IoT	Internet of Things
IoUT	Internet of Underwater Things
IoVT	Internet of Vehicular Things
LBC	Lattice-Based Cryptography
LWE	Learning With Error
M2M	Machine to Machine
MIMO	Multiple Input Multiple Output
MITM	Man in the Middle
MPKC	Multivariate Public Key Cryptography
NDN	Named Data Networking
NOMA	Non-Orthogonal Multiple Access
NTRU	Nth Truncated Ring Polynomial Unit
OFDM	Orthogonal Frequency Division Multiplexing
OTP	One Time Password
PKC	Public Key Cryptography
PUF	Physically Unclonable Function
RFA	Reverse Frequency Allocation
RFID	Radio-Frequency IDentification
RSA	Revest Shamir Adleman
SDN	Software Defined Networking
SKC	Symmetric Key Cryptographic
SS	Secret Sharing
SSL	Secure Socket Layer
SVP	Shortest Vector Problem
TLS	Transport Layer Security
V2V	Vehicle to Vehicle
Wi-Fi	Wireless Fidelity

- **Hybrid Schemes:** Hybrid schemes combining both pre and post quantum cryptography for efficient implementation and backward compatibility is yet to be standardised.
- **Risk Assessment Frameworks and Mitigation Measures:** Thorough risk analysis and proactive steps to guarantee the resilience and reliability of 5G IoT security structures should be the default requirement of any solution.
- **AI Integration:** Although AI is being integrated in a big way, but several challenges still remain such as the availability of 5G Specific dataset, model security, transparency through XAI and use of pretrained model.
- **Real Time Requirements:** Many applications and core operations of 5G IoT have real time constraints. Design

of PQC algorithms with complex computations running on resource constrained IoT devices for real time applications is hard to achieve.

Several researchers have conducted surveys on different aspects that highlight the limitations of IoT networks due to differences in network devices and limited computational power. Some researchers also portray intelligent attacks and countermeasures based on machine learning and deep learning approaches. Scholarly research articles like [22], [23], [24], [25], [26], [27], [28], [29], [30] emphasized on intelligent intrusion detection systems. However, we believe that a network must incorporate a certain level of security standard before its deployment. The knowledge of securing the exchange of data via various encryption techniques plays a vital and primary role before proceeding to an intrusion detection system. It is worthy to note that all public key cryptography tools do not fit appropriately in the IoT context owing to their limited computational power. Our foremost ambition is to list the public-key cryptography paradigms that are compatible with resource-constrained IoT applications. The advent of quantum computers has further revealed the loopholes in traditional public key cryptography, paving the way for post-quantum security schemes. In the cryptographic literature, terms such as “post-quantum”, “quantum-resistant”, “quantum-safe”, and “quantum-effective” are often used, sometimes interchangeably, though they are not always clearly defined. Post-quantum generally refers to cryptographic algorithms that are intended to remain secure in the presence of quantum adversaries whereas Quantum-resistant typically suggests that a scheme is believed to be hard to break using both classical and quantum computers, although this belief lacks formal proof. The term quantum-safe is sometimes used to convey a stronger notion of long-term security in a quantum future, but its use is largely heuristic and not grounded in rigorous guarantees. Quantum-effective, though less commonly used, describes scenarios or attacks where quantum algorithms provide a computational advantage. As far as we have explored, we did not find a single survey detailing both approaches in a unified manner. We compare the existing works with our study in Table 2. The following convention is followed in Table 2 to assess the exploration of earlier studies:

- ✗ denotes that the attribute is completely unaddressed
- ✓ represents that the attribute is addressed however, does not meet a desirable level or requires more emphasis
- ✓✓ indicates that the attribute is completely addressed

The limitations of earlier studies motivate us to pen this survey with the main objective of providing a contrast between the traditional pre-quantum and post-quantum security schemes for 5G-enabled IoT. Our major contributions include the following aspects:

- 1) We provide a systematic and brief review of state-of-the-art 5G-enabled IoT security schemes while emphasizing the primary contributions of our study in comparison to the previous surveys.

TABLE 2. Comparative analysis of state-of-the-art surveys at a glance.

Survey Reference	Year	Primary Objective	Exploration of		Security Solutions explored		5G IoT Security Challenges	Integration of Digital Twin - IoT	Integration Challenges	Risk Assessment and Mitigation
			Attacks	Applications	Traditional PKC	Post Quantum PKC				
Fernandez [31]	2019	Transition from pre to post-quantum security	X	X	X	✓	X	X	X	X
Hassan et al. [32]	2019	Security attacks and countermeasures	✓	X	✓	X	X	X	X	X
Roy et al. [33]	2019	Post-quantum Cryptographic prelims for IoT	X	X	X	✓	X	X	X	X
Gunathilake et al. [34]	2020	Lightweight Security Solutions for IoT	X	X	✓	✓	X	X	X	X
Shim et al. [35]	2021	Post-quantum solutions for IoVT	X	X	X	✓	X	X	X	X
Malina et al. [36]	2021	Post-quantum solutions for IoVT	✓	✓	X	✓	X	X	X	X
Chawla et al. [37]	2021	Technology shifts from classical cryptographic techniques to 5G-enabled IoT	✓	✓	✓	✓	✓	X	X	X
Seyhan et al. [38]	2022	Lattice-based solutions for IoT Security	✓	X	X	✓	X	X	X	X
Iqbal et al. [39]	2023	Post-quantum Security Solutions	✓	X	X	✓	X	X	X	X
Awouda et al. [40]	2024	Integrated framework for IoT and Digital twin	X	X	X	X	X	✓	X	X
He et al. [41]	2025	5G Security and Privacy with AI	✓	X	X	X	✓	X	X	X
Rishiwal et al. [42]	2025	Machine learning and Quantum Computing for IoT	✓	X	X	✓	X	X	X	X
Wang et al. [43]	2025	Post quantum cryptography in Blockchain and IoT	✓	✓	X	✓	X	X	X	X
Mansoor et al. [44]	2025	PQC for authentication and privacy in IoT	✓	✓	X	✓	X	X	X	X
This study	-	Pre- and Post-Quantum Security aspects for 5G-enabled IoT domain	✓	✓	✓	✓	✓	✓	✓	✓

- 2) We summarize the emerging trend of 5G and Digital twin and emphasize their impact in the context of IoT Applications.
- 3) We provide a detailed review of attacks and cryptographic solutions proposed as a countermeasure while analyzing their feasibility in practical implementation and computational and communication overheads.
- 4) We add a new classification of solutions: traditional public key schemes and post-quantum public key schemes to secure the 5G-enabled IoT subsystems and provide a contrast between the two approaches.

B. COMPOSITION OF THE SURVEY

We value the time of our readers and have conscientiously crafted this survey by following an organized and systematic procedure. In the initial phases, we consulted renowned journals' databases such as IEEE, Elsevier, Wiley, ACM Digital Library, Springer, Science Direct, Google Scholar, etc., to gather articles matching keywords like "Secured IoT", "5G and IoT", "IoT Security Protocols", "5G and Post-quantum in IoT", etc. The initial list of articles revealed that most schemes in the literature are based on RSA, ECC, or lattice-based cryptography. This guided us to refine our search queries to "5G RSA IoT", "5G ECC IoT", "5G IoT DH", "5G Secret Sharing IoT", "5G Lattice-Based Approaches for IoT", "5G Code-Based IoT", "5G IoT Applications on LWE", etc. A visual representation is provided in Figure 2.

The number of research articles found included journals, conference proceedings, chapters, etc., with a focus on journal and conference proceedings. To keep our readers abreast of current research, we limited our survey to the recent past years (2020–2025) and

explored innovative schemes proposed by scholars in the IoT security domain.

1) SELECTION CRITERION

Given the large pool of articles, we devised the following criterion to filter out redundant articles:

a) Identification and Formation of **Research Questions**:

- RQ:1 What are the unique vulnerabilities of 5G-enabled IoT applications?
- RQ:2 What are the prevalent attacks in IoT networks and their impact on organization or end-user privacy?
- RQ:3 What solutions exist in the literature and how do they balance theoretical analysis with practical efficiency in resource- constrained environments?
- RQ:4 How do quantum computers disrupt traditional public key crypto-systems?
- RQ:5 What post-quantum solutions exist and how practical are they in 5G-enabled IoT environments?
- RQ:6 How can an appropriate crypto-system be determined for a given application with predefined constraints?

These questions guided our search for articles that addressed these gaps.

b) Must meet the **Quality Assessment Points**, including but not limited to:

- QA:1 Does the article establish a *novel approach* to ensure security in IoT networks?
- QA:2 How does the proposed scheme *address privacy or security threats* in 5G-enabled IoT networks?

- QA:3 To what extent is the devised paradigm *practical* in the context of 5G-enabled IoT applications?
- QA:4 Is the scheme *lightweight* enough to operate on resource-constrained applications?
- QA:5 Does the scheme mitigate threats posed by the *advent of Quantum Computers*?
- QA:6 Does the scheme provide a sufficient security level within an acceptable *communication delay*?
- QA:7 How efficient is the solution in terms of *computational cost* regarding running time and space usage?

These standard approaches helped us to present a comprehensive survey with the clear intention of offering a thorough understanding of the current state of research and the progress made in strengthening security mechanisms in IoT networks. We have surveyed and reported on 90 articles. Table 3 enlists the number of articles considered at each stage.¹

TABLE 3. Number of articles under consideration at each stage of the selection strategy.

Source	Stage 1	Stage 2	Stage 3	Stage 4
IEEE Xplore	3002	865	165	37
Elsevier	1123	560	67	20
ACM Digital Library	793	400	98	10
Springer	4207	1256	300	12
Wiley	300	70	15	1
MDPI	207	103	23	5
Miscellaneous	90	34	20	5
Total Count	10111	3650	785	90

C. ORGANIZATION OF THE PAPER

The remainder of the paper is organized in the following fashion: We briefly point out the impacts of integration of 5G with IoT in Section II. Sections III and IV detail the security breaches and security requirements in the IoT network. The taxonomy of the proposed solutions is presented in Section V. Section VI and Section VII review the current state of research in traditional public key cryptography and post quantum public key cryptography. Open Research Directions are discussed in Section VIII. We conclude the survey in Section IX.

II. IMPACT OF 5G ON IoT

While many current IoT devices rely on 3G or 4G-LTE for connectivity due to their modest data requirements, a new generation of IoT devices demands higher throughput rates. As these requirements push the boundaries of existing wireless technologies, 5G emerges as a promising solution to meet these demands effectively [45]. With advancements in wireless communication technologies, 5G is becoming more and more real and has the potential to completely change the Internet of Things (IoT) landscape. Traditional IoT applications like Smart Homes, industries, and farming typically demand minimal wireless capabilities due to the limited processing power of the devices involved. However, emerging trends such as Smart Cities and IoVT necessitate

significantly higher throughput rates to deliver optimal user experiences, a capability uniquely provided by 5G networks [3], [45], [46]. In essence, the advent of 5G technology addresses the evolving connectivity requirements of IoT devices, enabling the realization of more sophisticated and data-intensive applications across various sectors. As IoT becomes increasingly pervasive in everyday life, the scalability and flexibility offered by cloud computing will be essential for managing the vast amounts of data generated by IoT ecosystems. The major additions that 5G brings in are highlighted in Figure 3 and listed as follows:

- 1) Multiple Input Multiple Output (MIMO): With the proliferation of IoT devices, which often require low-power, low-latency connectivity, Massive MIMO can efficiently handle the simultaneous transmission and reception of signals from numerous IoT devices within a given area. This enables better utilization of the available spectrum and improved coverage, crucial for supporting the massive connectivity demands of IoT deployments [3], [37], [46]
- 2) Non-Orthogonal Multiple Access (NOMA): NOMA's ability to support massive high-speed connectivity with low latency makes it particularly well-suited for IoT applications where devices need to exchange data quickly and reliably. In IoT scenarios, where numerous devices may have varying data requirements, NOMA's flexibility in resource allocation can ensure efficient use of the network resources, optimizing connectivity for diverse IoT devices [37], [46].
- 3) Millimeter Wave (mm-Wave): While millimeter wave technology offers increased spectrum bandwidth and freedom, it also presents opportunities for IoT applications. The higher bandwidth available in millimeter wave frequencies can accommodate the large data volumes generated by IoT devices, such as those used in industrial IoT (IIoT) environments. Additionally, the shorter wavelengths enable more precise localization and tracking of IoT devices, enhancing their usability in various IoT use cases, including asset tracking and monitoring [37].

Despite several positive changes, 5G also brings in a new set of challenges associated with implementation, security, interference etc. They are described as follows:

- 1) Interference Challenges: 5G faces technical complexities, particularly with interference sensitivity, where even mild rain in urban settings can cause disruptions. Despite mitigation efforts, widespread adoption requires addressing interference issues through more extensive architectures, as evidenced by small cell tests in various countries [47], [48]. Inter-channel interference arises from the close proximity of devices operating on different frequencies, leading to interference when high-power signals are transmitted over short distances. Additionally, interference can occur from the utilization of heterogeneous networks (hetnets) with Orthogonal Frequency Division Multiplexing

¹Figures are indicative and may differ by 5–6 in count.



FIGURE 2. Process of selection of appropriate articles.

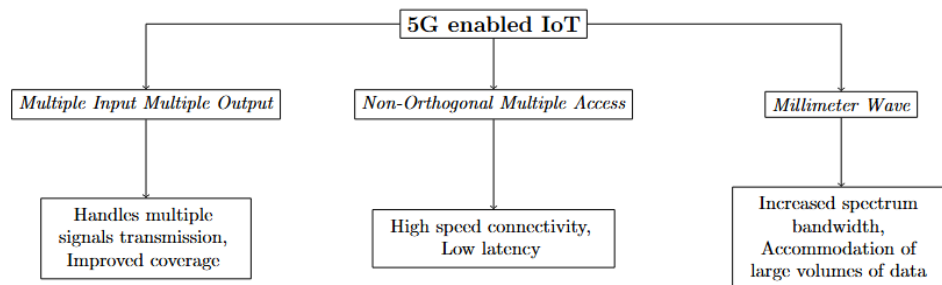


FIGURE 3. Impact of 5G communication in IoT.

(OFDM). Solutions for reducing Inter-channel interference include employing reverse frequency allocation (RFA), which is a proactive approach to mitigate interference [3].

- 2) Unequal Coverage and Urban Focus: The costly nature of 5G architectures currently prioritizes urban areas, leaving rural regions under-served. This socio-economic divide hinders the promised benefits of improved connectivity for disadvantaged communities, reflecting ethical and social implications.
- 3) Ethical Dilemmas of IoVT: The integration of 5G with IoVT technologies raises significant ethical concerns, especially with the rise of autonomous driving. While 5G facilitates precise vehicle control, it also amplifies moral dilemmas in cases of accidents involving autonomous vehicles, highlighting the need for ethical frameworks in technology deployment.
- 4) Security Risks: Connecting vehicles to the internet poses security threats, such as remote hijacking by hackers, leading to potential scenarios of vehicle-based kidnappings. This highlights the imperative for robust cybersecurity measures in the development and implementation of connected vehicle technologies.
- 5) Cost Barriers in 5G-enabled IoT: The implementation and utilization costs associated with 5G infrastructure and services may pose significant financial obstacles for many IoT applications, particularly those requiring large-scale deployment. Upgrading existing IoT devices to be compatible with 5G networks can further

strain budgets, potentially limiting the adoption of 5G-enabled solutions across various sectors.

- 6) Compatibility Challenges with 5G Networks: Certain IoT devices may lack compatibility with 5G networks, restricting their functionality and usefulness in specific applications. This compatibility gap could impede the seamless integration of IoT technologies with 5G infrastructure, potentially limiting the scope and effectiveness of IoT deployments reliant on advanced 5G connectivity.

The Internet of Things (IoT) is expected to undergo a significant transformation with the arrival of 5G. With low latency, ultra-fast data transmission, and seamless connectivity across a wide range of devices, 5G-enabled IoT is ready to serve to diverse application domains. Each smart application may offer a variety of services to enable a more intuitive environment. However, we are nowhere near exhausting the potential applications of IoT. The Internet of Things permits the combination of sensing, actuating, communication, routing, authentication, identity, and computation, as well as the provision of multiple services on demand, allowing access to the information of any smart gadget at any time. We have summarised the key applications in Table 4.

III. SECURITY BREACHES IN 5G-ENABLED IoT NETWORK

Users are becoming more dependent on IoT services thanks to the exponential proliferation of IoT devices. It is quite hard to imagine a world without technology. Attackers and intruders

TABLE 4. Summary of 5G enabled IoT applications.

Application domain	Key components	Use cases	Recent works	IoT integration benefits	Key Challenges
Internet of Medical Things (IoMT)	Sensing and transmission of patient attributes, automated diagnosis	Remote patient monitoring, robotic surgery, medical image analysis	[49]–[51]	Accessibility of healthcare, faster and accurate diagnosis, availability of patient history	Highly sensitive data, absolute reliability of devices
Internet of Underwater Things (IoUT)	Autonomous underwater vehicles (AUV), sensors and other devices transmit underwater parameters to surface unit	Underwater activity and security monitoring, oil and gas exploration, supervision of pollution and marine life	[52]–[54]	Accurate and reliable information about under water system	Acoustic signals, Latency, limited bandwidth, localization, energy harvesting
Internet of Vehicular Things (IoVT)	Communication between Road side unit and Onboard unit	Autonomous driving, traffic management	[55], [56]	Traffic efficiency and safety, vehicle maintenance	Privacy preservation, real time operation, access control
Digital Twin	Virtualization of real world data and object through automated exchange of data and control information	Digital Twin of production line, smart grid	[40], [57]–[59]	Monitoring, predicting, and enhancing performance in complex systems and processes	Real time data integration, data quality, model accuracy
Smart Home	Central monitoring point for all the networked sensors and appliances	Smart lock, smart refrigerator	[60], [61]	Energy efficiency, convenient and remote control, automation	Privacy, security, heterogeneity
Smart Grid	Sensors for grid operation monitoring, protection and control	Siemens Grid Software Suite, electricity theft detection	[62]–[64]	Real time estimation of demand and supply, predictive maintenance, faster fault detection, increased transparency	Reliability and resilience, scalability, grid stability
Supply Chain	Sensors and equipment to track and transmit key parameters	Fake product identification, HVAC system monitoring for perishable products	[1], [65]	Real time tracking, activity based payment, key insights on bottlenecks and future demands	Privacy, scalability, interoperability, regulatory compliance
Precision Agriculture	Sensors to measure soil moisture, minerals etc, AI based prediction, automation	Automated disease detection, Crop suggestion	[66], [67]	Enhanced productivity and quality, optimize resource utilization e.g. water	Node tampering, prediction accuracy

keenly look for opportunities to exploit the network covertly in order to steal sensitive information or cause financial losses to a business, individual, or organization. They have found the reliability of the devices to be an attractive target because of their deployment in remote locations out of the user's line of sight. Attackers can viably cause physical damage to the edge devices, introduce harmful software (malware) to compromise the system, or steal personal data, including identities, resulting in privacy breaches. IoT devices are resource-constrained applications. Their peculiar characteristics also poses certain challenges:

- 1) Large number of IoT devices: The evolution of 5G has increased the number of devices that can be connected per area. Since the number of devices connected to IoT is increasing by leaps and bounds, the task of assigning unique addresses to each device is a major cause of concern.
- 2) Heterogeneity of Network Devices: Different working paradigms of applications in IoT networks pose challenges of interoperability and scalability.
- 3) Limited Power: IoT devices are mostly battery-operated and have limited energy and power. IoT devices should be compact and consume low power.
- 4) Self-adaptability: Devices should be smart enough to operate in various environments. For instance, cameras need to be intelligent enough to recognize that while

flash may be prevented during the day, it is unavoidable at night.

- 5) Self Configuring: IoT devices may update their software with the least amount of user interaction in order to comply with standards. Additionally, they must be capable of performing dynamic operations like the adoption of new devices to an existing network and dynamic routing. In a vehicular network, for example, automobiles are seen as mobile nodes. As 5G offers a higher data rate, these nodes enter and leave the network in just a few milliseconds, such gadgets need to respond as quickly as possible.
- 6) Embedded Sensors and Actuators: IoT devices may gather data about the environment, process that data, and take action based on the results thanks to sensors and actuators working together. As a result, many different procedures and jobs may be automated, including home automation, energy management, and preventative maintenance. Processing algorithms must be as simple as possible and compatible with their limited computational abilities.

Many researchers propose different taxonomy of attacks possible in IoT systems. Hassan et al. [32], [68] proposed a taxonomy of attacks based on the vulnerability of a layer to an attack. Bout et al. [69] classified the attacks into four classes: Physical, Network, Software and Encryption

attacks. Alaba et al. [70] presented a new classification: application domain, architectural domain, communication channel, and data domain. Frequent classifications based on target layers are also available. Most prominently, cyber attacks are possible in IoT systems when:

- 1) Sink node communicates with cloud server
- 2) Users requesting services from cloud servers

Secure transmission of data in IoT is of paramount importance due to the sensitive nature of the applications involved. Confidentiality, integrity and authenticity of the data must be preserved using various cryptographic tools. Several efficient and lightweight schemes have been proposed to protect data in transit. In [71] and [72] authors have proposed a lightweight secure data transmission scheme with Elliptic Curve Menezes-Qu-Vanstone asymmetric key exchange and symmetric encryption of data. The performance of the scheme is verified through simulation in terms of computation and communication costs. Scalability of the system is ensured by dynamically changing the cluster size and security analysis confirmed the resilience of the system against MITM, brute-force and replay attacks. Harbi et al. have introduced a novel lightweight authentication and encryption scheme for cluster based WSN [73]. Cluster heads aggregate data from sensors and encrypt using ECC keys. Encrypted data are then signed using pre-shared symmetric keys and forwarded to base station which performs the computation intensive operations. Their scheme is designed to be resistant against brute-force, sinkhole and replay attacks and efficient with respect to computation and communication cost. A mutual authentication and secure data transmission scheme for heterogeneous IoT devices was presented in [74]. ECC, PUF, Hashing and ExOr operations were used to design a mutual authentication followed by session key establishment scheme between IoT devices, Network operator, IoT service provider and IoT users. Heterogeneity is introduced in the architecture by considering both resource rich and resource constrained IoT devices where constrained devices are benefited by the services of resource rich devices such as computation power. The scheme supports privacy, integrity, key escrow resilience, end to end security, forward secrecy as security features and has comparable computation and communication overhead. Lightweight, scalable and secure data transmission schemes are actively explored by researchers for diverse communication channels, protocols and devices of IoT.

Without fundamental elements like trust and privacy, an IoT system loses its effectiveness. Moreover, the privacy of data may face heightened risks with the increasing ubiquity of quantum computers. This underscores the need for a unique approach capable of withstanding quantum attacks. Classical attacks of Denial of Service, Man in the middle, Eavesdropping, Privacy attack, Data masquerading or impersonation, Replay attacks, Forward and backward secrecy attack, Password disclosure attack, Repudiation and Timing attacks are very common in an IoT system. Figure 4 presents a taxonomy illustrating these attacks in IoT.

IV. SECURITY REQUIREMENTS IN 5G-ENABLED IoT NETWORK

In convention, a secure system refers to the CIA triad: confidentiality, integrity, and availability [75]. The definition of each of these is as follows:

- 1) Confidentiality: It restricts the access of data to only authorized users.
- 2) Integrity: It validates that the data has not been altered during the course of transmission.
- 3) Availability: It ensures that the legal user has access to the services and obtains the data in its true form without any modifications.

IoT applications are task-specific and are oriented to sense some attribute of the surroundings in consideration. The need for secure systems is essential in 5G-enabled IoT applications in order to avoid security threats. Ensuring security standards is essential for each application. In brief, we discuss the need for a security framework with respect to each of the applications.

- 1) Internet of Medical Things (IoMT): An attack on the medical records of a patient may create confidentiality issues, resulting in a lack of trust between the patient and hospital authorities. Disclosure of the personal health records of patients may pose life-threatening risks.
- 2) Internet of Vehicular Things (IoVT): Privacy preservation and authorization are extremely crucial in IoVT. An attacker in the network may track the routes of others in the network or broadcast false updates pertaining to routes that cause road traffic.
- 3) Internet of Underwater Things (IoUT): Electric signals are subject to being tapped and can be re-engineered to extract or eavesdrop on information. Attackers may also inject malware and spyware to disrupt the costly setup.
- 4) Smart Home: An impersonation attack may provide access to thieves and help them plan a conspiracy against the users.
- 5) Smart grid: Smart grids are threatened by DoS attacks, denying services to people whose electrical needs are met by the smart grid.
- 6) Supply Chain: Privacy attacks are prevalent, leading to dent in brand reputation and consumer trust.
- 7) Precision Agriculture: Sensors are prone to physical node tampering. Corrupt reading and wrong predictions are certain challenges in smart agriculture.

From the above mentioned threats (as depicted in Table 5), it is evident that 5G-empowered IoT systems must maintain a standard security and meet the following requirements [75], [94], [95]:

- 1) Authentication
- 2) Access control
- 3) Confidentiality
- 4) Integrity
- 5) Privacy Preservation
- 6) Trust Management

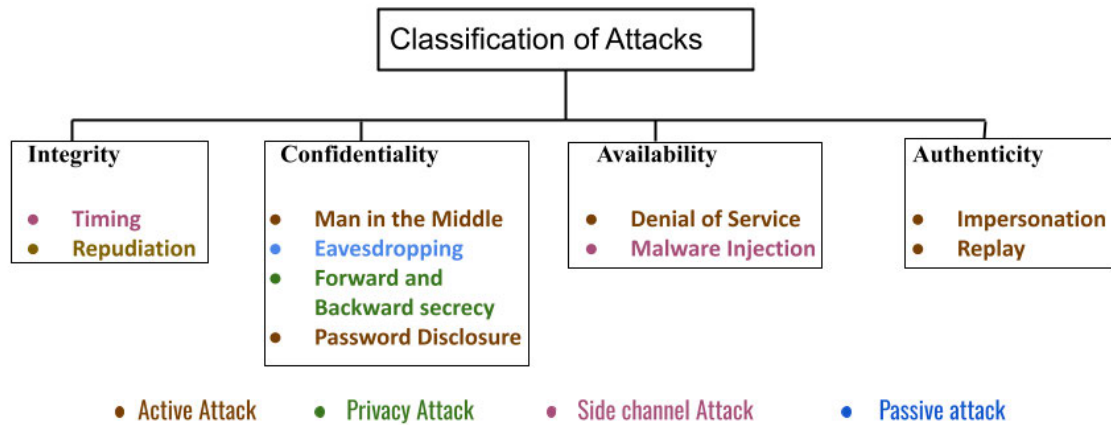


FIGURE 4. Attacks in IoT Network.

TABLE 5. Summary of security risks and challenges.

Application	Attacks susceptible to											
	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
IoMT	[76], [77] [78]	[77], [76] [78]	[76], [77]	[77], [76] [78]			[76]		[78], [79]	[77]	[76]	
IoVT	[80], [81]		[82]	[80]	[80]	[80], [82]	[80]	[81]		[81]	[80]	[80]
IoUT	[52], [53]						[83]		[52]	[83]		[83]
Smart Home	[84], [85]	[85]	[86]	[86], [84] [87]	[86]			[84]	[86]	[84]		
Smart Grid	[88], [89]	[89]			[89]	[89]	[88]			[89]	[88]	[89]
Supply Chain	[90], [91]				[91]	[90]	[90]	[91], [90]	[90]	[91], [90]		[91]
Precision Agriculture	[92], [93]					[93]		[92]	[92]	[92]		[93]

A1:Denial of Service, A2:Man in the Middle, A3:Eavesdropping, A4:Privacy attack, A5:Threat of quantum computers, A6:Impersonation, A7:Replay attacks, A8:Forward and backward secrecy attack, A9:Password disclosure attack, A10:Injection of Malware, A11:Repudiation, A12:Timing Attack

- 7) Availability
- 8) Resilience against quantum threats

To ensure these standards in the 5G-empowered network, many researchers have carried out careful experiments and proposed cryptographic solutions. We explore the potential solutions in the subsequent sections.

V. TAXONOMY OF CRYPTOGRAPHY SOLUTIONS AGAINST SECURITY BREACHES IN 5G-ENABLED IoT SYSTEMS

Many potential solutions have been proposed in the field of 5G-empowered security, which requires a systematic organization for acquiring meaningful insights from these diverse research components. In the past, many researchers, [22], [23], [24], [25], [26], [27], [28], [29], [30] presented a taxonomy of solutions based on machine learning and deep learning-based approaches. These surveys did not highlight any cryptographic solutions and are limited to intrusion detection systems.

Several contributions have been proposed using cryptography to meet the aforementioned standards in Section IV. We provide a new classification of available solutions based on their properties pertaining to quantum effectiveness.

In the present survey, we classify the whole space of available solutions into two broad classes, i.e., symmetric key cryptography (SKC) and asymmetric key cryptography (AKC) as shown in Figure 5. Owing to the inefficiencies in SKC, we have not explored this approach at length. The

prime contribution of this survey is to point out the potential solutions reported using the notion of AKC.

A public-key cryptography system is one that uses asymmetric keys for encryption and decryption. A pair of keys called the public key and the private key are mathematically computed. By the term mathematically, we mean that security of the keys are based on mathematically hard problems.

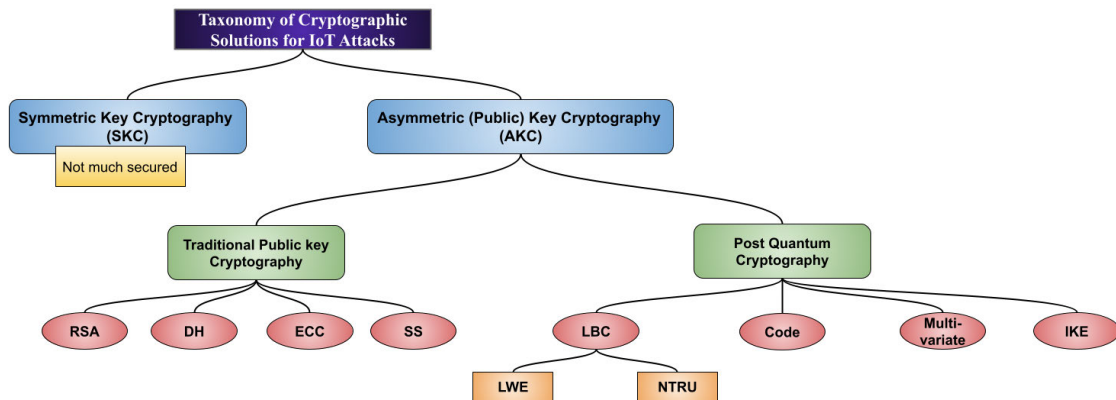
In asymmetric key cryptography, two keys are involved: a public key and a private key. A public key can be publicly published; however, the private key remains confidential to the creator. In AKC, first the receiver publishes the public key parameters. The sender encrypts the message with the public key published by the receiver. This encrypted message is termed as the cipher text and can be decrypted only with the private key of the receiver.

In the subsequent sections VI and VII, we broadly classified the solutions into two categories:

- 1) Traditional Public key Cryptography mainly detailing on DH, RSA, ECC and Secret Sharing
- 2) Post-quantum Public key Cryptography exploring the solutions reported on LBC, Code-based and Multivariate Polynomials.

Interpretation of Evaluation Criteria Now, we outline the guiding principles and assumptions used in our classification in tables 6, 7, 8, 9, and 11.

- **Lightweight:** A scheme is marked as lightweight if it is explicitly designed for constrained environments (e.g., IoT/embedded systems) and demonstrates favourable CPU usage, memory consumption, or energy efficiency



Abbreviations :

RSA : Revest Shamir Adleman; **DH** : Deffie Hellman; **ECC** : Elliptic Curve Cryptography; **SS** : Secret Sharing; **LBC** : Lattice Based Cryptography; **LWE** : Learning With Error; **NTRU** : Nth Truncated Ring Polynomial Unit; **IKE** : Isogeny Key Exchange

FIGURE 5. Taxonomy of cryptographic solutions for IoT Attacks proposed in recent past.

compared to conventional counterparts. Where exact metrics are unavailable, we rely on author claims backed by implementation or simulation.

- **Secure:** Security is determined based on formal claims of resistance to known attacks. Where available, we note the presence of standard security proofs such as IND-CPA, IND-CCA, or ECDLP resistance. Absence of such proofs is flagged where relevant.
- **Quantum-resistant:** A scheme is deemed quantum-resistant if it is based on hard problems believed to resist quantum adversaries (e.g., lattice problems, code-based, multivariate polynomial systems) as per NIST standards or equivalent references.

VI. TRADITIONAL PUBLIC KEY CRYPTOGRAPHY

This section is devoted to discuss the public key algorithms introduced before the advent of quantum computers. Several benchmark algorithms like Diffie-Hellman (DH) [96], RSA [97], and Elliptic Curve Cryptography (ECC) [98] are frequently used in the context of IoT. Secret sharing schemes are often used to provide authentication in resource constrained applications.

A. DIFFIE-HELLMAN (DH)

The Diffie-Hellman (DH) key exchange protocol, introduced by Whitfield Diffie and Martin Hellman in 1976 [96], represents the first public-key cryptographic system. This groundbreaking technique introduced the concept of public and private keys, laying the foundation for modern cryptographic communication. Since its inception, numerous adaptations and enhancements of the DH protocol have been proposed, particularly to address the challenges posed by resource-constrained environments such as the Internet of Things (IoT). In this section, we summarize several notable contributions that extend or improve the Diffie-Hellman framework in such settings.

Kim et al. [99] conducted a formal security analysis of the lightweight Ephemeral Diffie-Hellman over COSE (EDHOC) protocol, which operates at the application layer of IoT networks. Their study identified several vulnerabilities in the EDHOC protocol, including threats to message integrity and partial privacy leakage. To address these issues, the authors proposed enhancements that ensure the authentication of the initial message and the verification of the second message in the exchange. These modifications help mitigate resource exhaustion attacks while maintaining the protocol's lightweight nature.

To improve quantum resistance and security within constrained environments, Kanwal et al. [100] introduced a non-commutative variant of the DH protocol. Their scheme is built upon a ring of polynomials and relies on the hardness of a generalized decomposition problem in non-commutative algebraic structures, which is believed to be computationally infeasible to solve in polynomial time. The proposed protocol is resistant to various attacks, including man-in-the-middle, insider, and impersonation attacks, while also achieving perfect forward secrecy. Its lightweight design makes it particularly well-suited for IoT deployments where computational and energy resources are limited.

Nashwan et al. [101] developed a secure authentication scheme for smart irrigation systems, utilizing a one-way hash function derived from the DH key agreement. Their architecture encompasses agricultural IoT devices and gateway nodes, including authentication servers and field professionals. The protocol ensures mutual authentication and session key establishment while preserving perfect forward secrecy. Formal verification using Burrows-Abadi-Needham (BAN) logic demonstrates the scheme's robustness against a variety of threats, including de-synchronization, impersonation, replay, password guessing, man-in-the-middle, and privileged insider attacks.

Ruggeri et al. [102] presented an energy-efficient variant of the Triple Diffie-Hellman protocol tailored for IoT applications. Their proposed scheme allows IoT devices to obtain secure session keys by expending only the minimal energy required to transmit an HTTP request, with cryptographic computations offloaded to a remote server. This approach significantly reduces the energy burden on IoT nodes. Additionally, the study advocates the integration of blockchain technology to securely disseminate the established keys, enhancing both scalability and trustworthiness in decentralized IoT ecosystems.

Hassan et al. [103] proposed a hybrid scheme involving RSA and DH to prevent MIM and Reused key attack. Dual encryption, first by public key followed by private key, is employed under primitive roots and modular arithmetic to achieve forward secrecy and key revocation capabilities. The security of the scheme is verified in AVISPA under Random Oracle model and superiority of the scheme is established with respect to communication cost and impersonation probabilities against other similar works.

To solve the problem of single point of failure i.e. cloud server in X3DH, Reed Solomon Erasure encoding based key partitioning for distributed storage is proposed in [104]. The optimal storage nodes are determined using integer linear programming solver. They established the suitability of the system for mobile edge network using thorough evaluation of computation and storage overhead. The proposed scheme is claimed to have better success rate than greedy and random approaches.

To summarize, the Diffie-Hellman algorithm and its various adaptations have shown significant promise in securing IoT environments. From lightweight implementations like EDHOC to quantum-resistant non-commutative schemes and energy-efficient protocols leveraging blockchain, DH-based approaches continue to evolve to meet the unique constraints of IoT systems while ensuring secure key exchange and robust protection against diverse cyber threats.

Table 6 summarizes the advances on DH Schemes.

B. RIVEST-SHAMIR-ADLEMAN (RSA) ALGORITHM

The RSA algorithm, introduced by Rivest, Shamir, and Adleman in 1978 [97], is one of the most foundational and widely adopted asymmetric encryption systems in cryptography. Its security is based on the computational difficulty of factoring large integers, a task for which no polynomial-time classical algorithm is currently known [105]. RSA continues to find broad relevance in modern secure communication systems, particularly in the context of the Internet of Things (IoT), where lightweight, energy-efficient, and attack-resilient cryptographic mechanisms are essential. In this subsection, we survey the evolution and adaptation of RSA across various IoT applications, highlighting its effectiveness, optimizations, and hybrid integrations.

One of the early notable applications of RSA in IoT is the biometric-based authentication scheme proposed by

Bagha et al. [106]. Their system, RSA-B-ASH-S, combines password authentication, smartphone-based identification, and biometric verification to establish a secure three-factor authentication mechanism for smart home environments. The scheme is designed to be resilient to impersonation, replay, man-in-the-middle, and denial-of-service attacks, while also ensuring forward secrecy.

Addressing the resource limitations of IoT devices, Mondal et al. [107] implemented a hardware-optimized version of RSA using a Systolic-Montgomery multiplier. Their architecture improves computational efficiency while reducing power consumption and area overhead. Supporting key sizes up to 4K, the implementation is shown to be resistant to common modulus and timing attacks, making it suitable for high-security applications in constrained environments.

In a different direction, Vishwakarma et al. [108] applied RSA in the domain of image encryption for IoT systems. By combining RSA with Gaussian Pyramid (GP) decomposition and LAB color space processing, their scheme focuses on the luminance channel for efficient and secure image encryption. Preprocessing techniques such as median filtering and sharpening were employed to enhance performance under noisy conditions, with evaluation metrics confirming improved entropy and robustness against visual and statistical attacks.

RSA has also been adapted for secure power infrastructure. Ahmed et al. [109] designed a secure communication protocol for smart grid-connected inverters. By encrypting data exchanged between IoT devices and the energy grid using RSA, their solution strengthens the cyber-security of power trading systems. The system demonstrates resilience against sniffing and spoofing attacks and is tailored for use with SSL protocols, making it highly suitable for green energy IoT networks.

To protect sensitive medical data in the Internet of Medical Things (IoMT), Elamir et al. [110] introduced a hybrid cryptographic scheme that combines RSA with DNA encoding. The RSA-encrypted binary image is mapped to a DNA sequence using a public key-driven mapping technique. This fusion of classical and biological computing paradigms enhances the confidentiality and integrity of medical data transmissions, providing strong resistance to eavesdropping.

Several works have extended the classical RSA algorithm to improve its performance and security guarantees. Imam et al. [111] proposed an extended RSA variant called XRSA, which uses four large primes and introduces intermediate exponents and XOR-based transformations during encryption and decryption. Although computationally more demanding, XRSA offers significantly greater resistance to brute-force attacks. The authors suggest the use of parallelization to offset the added computational overhead in IoT environments.

In another hybrid approach, Prem et al. [112] proposed a blockchain-based IoT data transmission framework that integrates SHA-512-based authentication, XOR-RSA

TABLE 6. Recent advances in IoT Security based on Diffie-Hellman (Pre Quantum).

Reference	Year	Application	Type	Time	Evaluation metrics	Attack mitigated	Lightweight	Security Scheme
Kim et al. [99]	2021	-	-	-	-	Message exhaustion	Yes	DH (EDHOC)
Kanwal et.al [100]	2022	-	-	0.19 s	Computation time, Communication cost, Passes Overheads	MIM, Impersonation, Forward Secrecy	Yes	DH
Nashwan et al. [101]	2022	Smart Irrigation	-	0.05712 s	Running time, Storage size, Intercommunication cost	MIM, Replay, Insider Attacker Impersonation, Forward Secrecy, Password Guessing	Yes	One way hash - DH
Ruggeri et al. [102]	2022	Smart City	-	-	Energy Consumption	Masquerade	Yes	Blockchain - X3DH
Hassan et al. [103]	2025	-	-	0.67 ms	Communication cost, impersonation probability	MIM, Reused Key	Yes	RSA - DH
Altaweel et al. [104]	2025	Mobile Edge Network	-	9 ms	Communication overhead, storage overhead, Success rate	Denial of Service	No	X3DH

encryption, load balancing using BM-SSA, and secure storage on a distributed ledger. This multi-layered security model is designed to minimize latency, reduce packet loss, and provide energy efficiency—characteristics vital for real-time, low-power IoT applications.

Optimizing RSA performance through number-theoretic techniques, Kollipara et al. [113] employed the Chinese Remainder Theorem (CRT) to accelerate RSA key generation and encryption/decryption processes. Their three-module framework—comprising OTP-based authentication, lightweight RSA communication, and Feistel network encryption—provides robust protection against spoofing attacks and unauthorized access, particularly in low-powered IoT deployments.

Building further on hybrid schemes, Bhati et al. [114] combined RSA with DNA cryptography for secure cloud processing via IoT devices. Their method offers a two-layer encryption approach that ensures user authentication, privacy, and data integrity. Penetration testing confirms its robustness and effectiveness over traditional single-layer RSA implementations.

Odyuo et al. [115] presented a multifactor mutual authentication scheme based on device serialization and digital signatures. The authentication protocol ensures that both the IoT device and the server are validated before granting access, with the system restarting authentication upon failure. This strict verification mechanism enhances trust and security in distributed IoT networks.

Anusuya Devi and Sampradeepraj [116] introduce a hybrid encryption framework for Wireless Sensor Networks (WSNs) that combines AES (symmetric cryptography) for data encryption and RSA (asymmetric cryptography) for secure key distribution. Their two-phase model divides the plaintext into two parts, encrypting one with AES and the other with RSA, thereby optimizing performance and enhancing data security. This approach balances the computational efficiency of symmetric encryption with the secure key management advantages of asymmetric schemes. Evaluations show improved encryption/decryption time and total execution efficiency, making it a strong candidate for secure communication in resource-constrained environments like WSNs.

Chang et al. [117] proposed customized AES and RSA combined scheme for low powered devices. By reducing the number of rounds in AES and introducing three threaded computations, at the same time moving from dual to triple prime numbers for reduced bit length in RSA, the authors gained significant advantage in latency and energy consumption.

In summary, RSA continues to be a central cryptographic primitive in IoT security due to its strong theoretical foundation and adaptability. Recent research has focused on enhancing its computational efficiency, combining it with other cryptographic paradigms, and integrating it into layered security frameworks. These innovations make RSA well-suited to meet the growing demands of secure, scalable, and lightweight IoT ecosystems.

Table 7 summarizes the advances on RSA Schemes.

C. ELLIPTIC CURVE CRYPTOGRAPHY (ECC)

Elliptic Curve Cryptography (ECC) is a public-key cryptographic technique that leverages the mathematical structure of elliptic curves over finite fields. Unlike RSA, which is based on the hardness of integer factorization, ECC relies on the difficulty of the Elliptic Curve Discrete Logarithm Problem (ECDLP) [118]. One of the key advantages of ECC is that it offers equivalent levels of security with much smaller key sizes, making it an efficient and robust option, particularly suitable for environments with limited computational resources such as IoT. Scalar multiplication, the core operation in ECC, involves computing $Q = kP$ for a given point P on the curve and a scalar k , which is computationally infeasible to reverse for large values of k [98].

To tackle interoperability and security challenges in IoT networks, Sahoo et al. [119] proposed an ECC-based authentication and key agreement protocol that integrates AES, hash functions, and XOR operations. Their scheme was designed to operate efficiently in heterogeneous IoT environments. Building upon the lightweight nature of ECC, Hammi et al. [120] presented a scheme that incorporates the One-Time Password (OTP) mechanism to generate fresh keys for each communication exchange, outperforming standard

TABLE 7. Recent advances in IoT security based on RSA (Pre Quantum).

Reference	Year	Application	Type	Time	Evaluation metrics	Attack mitigated	Lightweight	Security Scheme
Bagha et al. [106]	2020	Smart Home	-	8 ms	Computation speed, Storage requirements, Communication Overheads	MIM, DoS, Replay, Impersonation, Forward Secrecy	No	DH; RSA
Mondal et al. [107]	2021	-	-	6.2 μ s	Energy Consumption	Common Time attack, Timing attack	Yes	RSA
Vishwakarma et al. [108]	2021	-	Image	-	Entropy, Histogram	Gaussian attack	No	RSA
Ahmed et al. [109]	2021	Smart Grid	-	0.04 s	Computation delay	Spoofing, Sniffing	No	RSA
Elamir et al. [110]	2022	Healthcare	Image	50 s	Entropy, Correlation, Signal to Noise Ratio	Eavesdropping	No	DNA-RSA
Imam et al. [111]	2022	-	-	47.32 ms	Computation Speed	Brute force	No	XOR-RSA
Prem et al. [112]	2022	-	-	5676 ms	Energy consumption, End to End delay, Packet loss, Throughput	Impersonation	No	XOR-RSA
Kollipara et al. [113]	2023	-	File	12 ms	Computation Speed	Spoofing, Impersonation	Yes	RSA;SIT
Bhati et al. [114]	2023	-	File	45 ms	Computation Speed	Eavesdropping	No	DNA-RSA
Odyuo et al. [115]	2023	-	-	-	-	Impersonation	No	RSA
Anusuya & Sampradeep [116]	2024	-	-	-	Security	DoS	Yes	RSA
Hassan et al. [103]	2025	-	-	0.67 ms	Impersonation probability, Communication cost	MIM, Reused key	Yes	RSA - DH
Chang et al. [117]	2025	-	LoRaWAN	5-15 ms	Security, Latency, Energy	MIM, Spoofing, Replay	Yes	RSA - AES

HOTP and TOTP methods in terms of computational and communication overhead.

Rostampour [121] identified vulnerabilities in existing hash-based authentication schemes and introduced ECCbAP, a protocol that excludes hash functions and provides stronger privacy guarantees for edge devices interacting with the cloud. Majumder et al. [122] improved the widely used Constrained Application Protocol (CoAP) by embedding ECC-based authentication. Their scheme, ECC-CoAP, ensures secure communication through untrusted channels and demonstrates formal security guarantees via the AVISPA tool and BAN logic.

Focusing on key generation, Aswathy et al. [123] proposed a multi-factor authentication scheme where ECC keys are generated using the linear congruential method, ensuring stronger randomness. For secure data hiding, Geethanjali et al. [124] integrated ECC with LSB steganography. Their four-phase approach combines encryption, steganographic insertion and extraction, and decryption to protect confidential IoT data.

In the context of fine-grained access control, Cheng et al. [125] developed a CP-ABE scheme based on ECC, replacing bilinear pairings with scalar multiplications to reduce overhead. Their model incorporates temporal and spatial attributes and distributes workloads across multiple servers to eliminate bottlenecks.

Yao et al. [78] addressed the security needs of IoMT by proposing an ECC-based scheme enhanced with Physically Unclonable Functions (PUFs). The scheme is evaluated using BAN logic and found to be efficient in both communication and computational aspects. Zhou et al. [79] introduced

RAKI, a robust ECC-based three-party authentication and key agreement protocol for medical IoT, validated using the random oracle model.

Selvi et al. [126] designed an ECC-reliant authentication scheme for secure cloud communication with smart IoT devices. Their framework outperforms prior models by resisting a wide range of attacks, including replay, traceability, integrity breaches, man-in-the-middle (MIM), and denial-of-service (DoS) attacks.

Ghaffar et al. [127] explored the use of PUF and chaotic map for mutual authentication in healthcare network. Unique PUF installed in mobile sensing devices prevent physical tampering and cloning attack. On the implementation front they developed a testbed using Raspberry Pi, FPGA and Google Colab. Comprehensive security and performance analysis of the scheme executed under RoR model and Scyther tool and the authors claimed to achieve 46.84% reduction in computational overhead and 31.3% reduction in communication cost.

Rexline et al. [128] presented certificateless signature based mutual authentication scheme combining ECC and one way generic hash for IoMT applications. The scheme reduced computation cost by 0.89 s and communication cost by 252 bytes and proved to be secure on Random Oracle model.

Yet another novel hybrid ECC and Zero knowledge proof based scheme is suggested in [129]. With the aim of privacy preserving authentication in sensitive IoMT communications, the authors have combined ECC and Fiat Shamir protocol and limited computation cost to 300 ms for 160 but security.

Lahraoui et al. presented an optimized version of ECC by reducing costly point multiplication operations with efficient

and reverse mapping operation of elliptic curves over prime fields [130]. With significant improvement in computation, communication and storage cost, the authentication scheme remains efficient even at higher security levels. Simulated evaluations and formal security analysis confirmed the security and performance of the scheme.

In summary, ECC has emerged as a leading cryptographic method for secure and lightweight communication in IoT ecosystems. Its ability to offer strong encryption with minimal resource consumption has made it a preferred choice in modern cryptographic applications across healthcare, smart devices, and cloud infrastructures.

Table 8 summarizes the advances on ECC Schemes.

D. SECRET SHARING SCHEMES

Shamir [131], [132] introduced the foundational (t, n) threshold secret sharing scheme, which has since become a cornerstone in cryptographic protocols. In this scheme, a secret $\mathcal{S}$ is divided among a group of participants $\mathcal{P}$ such that:

- No single share reveals any information about the secret.
 - Any t or more participants can collaboratively reconstruct the secret using Lagrange interpolation.
 - Fewer than t participants learn nothing about the secret.
- Each participant $\mathcal{P}_i$ receives a *share* computed as:

$$f(x) = \mathcal{S} + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1} \quad (1)$$

where $f(x)$ is a random polynomial of degree $t - 1$ and $\mathcal{S}$ is the constant term (i.e., the secret). The secret is reconstructed via Lagrangian interpolation:

$$\mathcal{S} = \sum_{i=0}^{t-1} f(x_i) \prod_{j=0, j \neq i}^{t-1} \frac{x_j}{x_j - x_i} \quad (2)$$

In large-scale IoT networks, the scalability and resource efficiency of key distribution are critical challenges. Tang et al. [133] addressed this by proposing a physical-layer key-sharing protocol for MIMO-based IoT devices. This method enables direct transmission of secret keys from sender to receiver, reducing latency and overhead.

Wang et al. [134] proposed LiPSG, a lightweight Q-learning framework for smart grids that incorporates additive secret sharing to preserve user data privacy. Here, the energy consumption data is divided into secret shares and securely transmitted to the control center, enhancing privacy in smart grid applications.

Sebbah et al. [135] developed a lightweight threshold-based secret sharing mechanism for MQTT-based IoT systems. Their scheme adds an encryption layer for secure data exchange, suitable for resource-constrained environments.

To address the limitations of message integrity in traditional secret sharing, Xiong et al. [136] proposed RSIS, which combines secret image sharing with extended Hamming codes. This technique embeds stego-images across cloud servers, enabling tamper detection and error correction.

However, it is limited to correcting single-bit errors and detecting only two-bit errors.

Tang et al. [137] enhanced security in IoT data outsourcing using a threshold secret sharing scheme combined with Software Defined Networking (SDN). Data is split into secret shares, bundled into packets, and securely routed by SDN-enabled routers, thus providing both efficiency and control.

Junejo et al. [138] introduced LoRa-LiSK, a shared key generation protocol for LoRa networks. It combines timestamp matching, statistical tests, signal filtering, and ECC-based error correction codes with SHA-2-based privacy amplification. The protocol achieves high key generation rates while maintaining low disagreement rates and low correlation in channel characteristics.

Hamad et al. [82] proposed SEEMQTT, a secure MQTT communication framework for IoVT using threshold secret sharing and trust delegation. Publishers define access policies, and KeyStores handle secret key generation using identity-based encryption to protect communication channels.

Wei et al. [139] proposed a chaotic compressed sensing-based image sharing method tailored for IoT. Their multi-group secret sharing protocol includes access-level-based participant division and chaotic masking for secure transmission. However, the scheme is constrained to transmit a single image and struggles with larger image sizes.

Subrahmanyam et al. [140] introduced a group key agreement protocol incorporating an Elliptic Curve Secret Sharing Scheme (ECSSS). The protocol is designed for distributed IoT environments and offers reduced computational overhead while enhancing security, making it suitable for lightweight applications.

To address the need for energy-efficient key generation at the physical layer, Tang et al. [141] proposed a group secret key generation method using wireless broadcasting. Their protocol eliminates the need for a secure channel, reduces energy consumption, and ensures forward secrecy.

Iwamura and Kamal [142] propose a user authentication framework that provides information-theoretic security using (k, n) -threshold secret sharing and two-party secure computation. Unlike conventional authentication systems reliant on public-key cryptography and vulnerable to quantum attacks, their approach offers quantum-resilient security with minimal computational overhead. The core innovation lies in eliminating the need for secure channels or a trusted third party (TTP) by employing secure computation to perform user verification and secure communication, even over insecure channels. Experimental results demonstrate the suitability of the proposed method for IoT environments, achieving execution times below 0.1 ms. This framework marks a pioneering step in applying secret sharing directly to two-party communication scenarios.

In [143] Bilinear Pairing and Shamir's Secret Sharing based scheme is proposed to address the multi party authorisation problem in IoT device access. After initial key establishment using Bilinear Pairing, threshold based

TABLE 8. Recent advances in IoT security based on ECC (Pre Quantum).

Reference	Year	Application	Type	Time	Evaluation metrics	Attack mitigated	Lightweight	Security Scheme
Sahoo et al. [119]	2020	-	-	17.3 ms	Security Strength, Computation and Communication Cost	Impersonation, Replay, Anonymity, Forward Secrecy	No	AES- ECC
Hammi et al. [120]	2020	-	-	108.75 ms μ s	Security Strength, Computation, and Communication Cost	Impersonation	Yes	ECC
Rostampour [121]	2020	-	-	-	Security Strength	MIM, DoS, Replay, Integrity, Traceability, Impersonation	Yes	ECC
Majumder et al. [122]	2021	-	-	28.77 ms	Security Strength, Communication, Computation Costs	MIM, DoS, Replay, Non-repudiation, Forward Secrecy, Impersonation, Password Guessing	No	ECC-CoAP
Aswathy et al. [123]	2021	-	-	44 ms	Computation, Communication overheads	Impersonation	Yes	ECC + Linear Congruential
Geethanjali et al. [124]	2021	-	Image	-	Mean Square Error, Peak Signal to Noise Ratio, Embedding Efficiency	Masquerading	No	ECC + LSB Steganography
Cheng et al. [125]	2021	Smart Grid	-	500 ms	Computation and Communication Cost	-	No	ECC Cp-ABE
Yao et al. [78]	2022	IoMT	-	10 ms	Security, Computation and Communication cost and Energy Consumption	MIM, Replay, Password Guessing, Forward Secrecy, Anonymity, Impersonation	Yes	ECC
Zhou et al. [79]	2022	IoMT	-	44.12 ms	Security, Computation and Communication cost	Password Guessing Traceability, Anonymity	No	ECC
Selvi et al. [126]	2023	-	-	-	Security, Computation and Communication cost and Energy Consumption	MIM, DoS, Replay, Integrity, Traceability, Impersonation	Yes	ECC
Ghaffar et al. [127]	2025	IoMT	-	0.1304 ms	Security, Computation and Communication cost and Energy Consumption	Impersonation, Physical capture, Ephemeral secret leakage	Yes	ECC+PUF+ Chaotic Map
Rexline and Anita. [128]	2025	IoMT	-	89.284 ms	Security, Computation, Communication and Storage cost	MIM, Tampering, Replay, Integrity, Impersonation	Yes	ECC+One way generic hash
Misra et al. [129]	2025	IoMT	-	300 ms	Computation cost	MIM	Yes	ECC+Zero Knowledge Proof
Lahraoui et al. [130]	2025	-	-	92.22 ms	Security, Computation, Communication and Storage cost	Impersonation, MIM, Replay	Yes	ECC

Shamir's Secret Sharing is used for authorisation of User request for IoT device access. An user request must be endorsed by a minimum threshold t numbers of authorisers before being accepted by the Gateway. Admin plays the role of dealer by distributing shares over a secure channel. The security of the scheme is formally verified in AVISPA and the scheme is found to be efficient in terms of timing and storage requirements. They have also developed a prototype of the system using Raspberry Pi, DHT sensors and an Android application.

In summary, secret sharing schemes are critical for secure, efficient key management in IoT systems. Modern adaptations enhance the original Shamir's framework with lightweight operations, error correction, identity-based encryption, and physical-layer implementations—offering scalable, secure communication solutions for increasingly interconnected IoT ecosystems.

Table 9 summarizes the advances in Secret Sharing Schemes.

VII. POST-QUANTUM PUBLIC KEY CRYPTOGRAPHY

Public key cryptography is an essential aspect of computer, network, and IoT security. However, recent advancements in computing technology pose threats to traditional cryptogra-

phy. For instance, in 2010, the 1024 bit RSA implementation was broken [144]. This has necessitated a substantial increase in the key size from 1024 bit to 2048 bit and 4096 bit [145]. Increasing the key size has an adverse effect on the computation cost and results in significant overhead.

The advent of super-fast quantum computers further challenged security measures. A quantum computer can now solve factorization problems that take classical computers exponentially long to solve [146]. As a result, exploring post-quantum security solutions becomes inevitable.

Mosca et al. [146] predicted that quantum computers might not be prevalent by the next 20 years; however, it is of utmost necessity to revolutionize our security system before their commercial emergence. There are different types of post-quantum public key cryptography solutions currently under development. These include: (i) lattice-based cryptography [147], (ii) code-based cryptography [148], (iii) multivariate-based cryptography [149]. We explore these techniques in the sub-sections following.

A. LATTICE-BASED CRYPTOGRAPHY

A lattice L is defined as a discrete set of points arranged in a periodic structure in an n -dimensional space. A set of linearly independent vectors, say $\{b_1, b_2, \dots, b_n\}$ forms

TABLE 9. Recent advances in IoT security based on secret sharing (Pre Quantum).

Reference	Year	Application	Type	Time	Evaluation metrics	Attack mitigated	Lightweight	Security Scheme
Tang et al. [133]	2020	-	-	-	Signal to Noise Ratio, Key bits error rate	Eavesdropping	No	Secret Sharing
Wang et al. [134]	2020	Smart Grid	-	2.16 ms	Computational Overhead	Eavesdropping, Masquerading	Yes	Q-learning Secret Sharing
Sebbah et al. [135]	2020	-	-	1.54 s	Computational Overhead	Eavesdropping, Masquerading	Yes	MOTT - Secret Sharing
Xiong et al. [136]	2021	IoMT	Image	-	Security strength, Image Quality, Peak Signal to Noise Ratio	Eavesdropping, Impersonation, 1-bit errors	No	Steganography - Secret Sharing
Tang et al. [137]	2021	Smart City	Image	-	Security strength, Communication overhead, Possibility of data compromise	Eavesdropping, Impersonation	No	SDN - (k, n) Secret Sharing
Junejo et al. [138]	2021	LoRaWAN	-	-	Security strength, Communication overhead, Entropy, Correlation	Eavesdropping, Masquerading	Yes	Secret Sharing
Hamad et al. [82]	2022	IoVT	-	1038 ms	Security strength, Communication Time	Eavesdropping, Impersonation	No	MQTT - Secret Sharing
Wei et al. [139]	2022	-	Image	-	Security strength, Peak Signal to Noise Ratio	Masquerading	No	Compression Sensing - Secret Sharing
Tang et al. [141]	2022	-	-	-	Security strength, Signal to Noise Ratio	Impersonation	Yes	Secret Sharing
Subrahmanyam et. al [140]	2022	-	-	-	Security strength, Cost of Operations, Computational overhead	Impersonation	Yes	Elliptic Curve Secret Sharing
Iwamura & Kamal [142]	2025	-	-	0.1 ms	Security strength	Impersonation	Yes	Secret Sharing
Choudhury et al. [143]	2025	-	-	581 ms	Computational overhead, Communication overhead, Formal Security Verification	Ownership transfer Device capture attack MitM and DoS	Yes	Shamir's Secret Sharing

the basis of lattice and lattice $L = \sum_{i=1}^n x_i b_i$ where x is a random number. Lattice-based cryptography is based on n -dimensional hard problems Shortest Vector Problem (SVP) and Closest Vector Problem (CVP) [150]. These problems still remain unsolved by quantum computers in polynomial time. According to [151], [152], and [153], the SVP problem can be solved in $2^{O(n)}$. N^{th} Truncated Ring Polynomial Unit (NTRU) and Learning With Error (LWE) are the two primary classes of lattice-based algorithms widely used in cryptography. The security of NTRU is based on hardness that is not proven to be reducible to solving the CVP in a lattice, whereas the security of LWE relies on the determination to be provably reducible to solving the SVP [150]. As a result, while NTRU offers less security, it is actually much more flexible and efficient in its implementation. In contrast, LWE provably immune to quantum attacks, but because of their relative inefficiency, many researchers have developed various effective schemes (for example, over rings, as in Ring-LWE) [150].

1) LEARNING WITH ERROR (LWE)

In 2009, Micciancio et al. [154] introduced the LWE, correlating it with the SVP problem. LWE algorithm distinguishes the set of vectors that are generated from an erroneous set of vectors derived from linear equations. LWE is a formulation to hide the secret within a set of erroneous equations. Choudhary et al. [155] proposed a classification of solutions

based on Ring LWE and Fully Homomorphic Encryption (FHE).

Saeed [156] presented an integrated approach involving LSTM based anomaly detection, SHA-256 based integrity verification, Q learning based automated response and LWE based quantum resistance. They optimized deployment of these computation intensive components using edge offloading for anomaly detection, lightweight hashing with salted input, incremental Q learning for faster convergence and precomputed key in LWE. Effectiveness of the scheme is verified through simulation environments.

The ongoing standardization of post-quantum cryptography (PQC) by NIST has selected CRYSTALS-Kyber as the key encapsulation mechanism (KEM) and CRYSTALS-Dilithium and FALCON as primary digital signature schemes. These algorithms are based on hard lattice problems and are designed to resist both classical and quantum attacks while offering practical efficiency across diverse computing platforms.

a: CRYSTALS-KYBER

Kyber is a lattice-based KEM whose security relies on the Module Learning With Errors (Module-LWE) problem over the ring $\mathbb{Z}_q[X]/(X^{256} + 1)$, with modulus $q = 3329$. It supports three security levels—Kyber512, Kyber768, and Kyber1024—roughly corresponding to AES-128, AES-192,

and AES-256, respectively. The Kyber KEM is constructed from a public-key encryption (PKE) scheme using the Fujisaki–Okamoto transform to ensure IND-CCA2 security. At its core, Kyber performs matrix-vector and vector-vector polynomial multiplications, with matrices and vectors composed of ring elements. Efficient computation is achieved through Number Theoretic Transform (NTT)-based polynomial multiplication, modular reduction via Montgomery and Barrett arithmetic, and compression techniques that reduce transmission overhead [157], [158].

The implementation of Kyber on ARMv7-based Cortex-A processors by Kim et al. [159] is the first of its kind. It leverages the NEON SIMD engine to parallelize NTT operations and proposes a modified Montgomery reduction using `vqdmulh` and `vhsb` instructions, avoiding costly long multiplications. Barrett reduction is optimized using $R = 2^{16}$ for implicit shifting, enabling parallel processing of 8 coefficients per vector register. Compared to PQM4 and vectorized reference implementations, the optimized implementation achieves up to **62%** improvement in NTT, **50%** in point multiplication, and **56%** in inverse NTT. For the complete Kyber768 scheme, improvements of **50%**, **43%**, and **52%** are reported in key generation, encapsulation, and decapsulation, respectively.

Zia et al. [160] further analyze Kyber on microcontroller-class devices such as ARM Cortex-M4, applying algorithmic and architectural optimizations including memory alignment, prefetching, loop unrolling, and use of scratchpads to mitigate memory latency. Their cross-platform evaluation reveals that the choice of optimizations is highly architecture-dependent, with some memory-focused optimizations yielding speedups of over $1.4\times$ on cache-sensitive platforms. They also show that despite Kyber's relatively larger ciphertexts compared to elliptic-curve cryptography (ECC), it outperforms ECC in both speed and energy efficiency on many embedded platforms, making it a viable replacement for secure communication protocols like TLS in IoT systems.

A novel application of Kyber in physical layer of wireless communication was presented in [161]. Two versions of Kyber were proposed by tweaking the Binomial Distribution random number generator, WKyberV1 as PKE and WKyberV2 as both PKE and KEM. They used 4QAM for signal encoding, BCH code for error correction and Additive White Gaussian Noise for channel distribution. Security analysis revealed the scheme to perform well with high SNR value signals and the scheme showed comparable spectral efficiency with BLE.

b: CRYSTALS-DILITHIUM

Dilithium is a lattice-based digital signature scheme, also based on Module-LWE and Module-SIS problems. It uses a larger modulus $q = 8380417$ and supports asymmetric matrix sizes for its three security levels: Dilithium2, Dilithium3, and Dilithium5. The signing process adopts the Fiat–Shamir with Aborts paradigm, where trial signatures are generated

and validated against predefined norm and hint constraints. Like Kyber, Dilithium relies on NTT-based polynomial multiplication, although it packs only 4 coefficients per 128-bit vector register due to the larger coefficient size [158].

Kim et al. [159] implement Dilithium with vectorization on ARMv7, merging NTT layers (using a 3-2-2 strategy) to minimize memory accesses. Their implementation achieves **54%**, **62%**, and **55%** improvement in NTT, point multiplication, and inverse NTT steps, respectively, over PQM4. End-to-end performance gains include up to **41%** in signing, and **14–17%** and **20–24%** in key generation and verification, respectively, across all parameter sets.

Complementing this, Zia et al. [160] explore implementation trade-offs across various embedded CPUs. They show that Dilithium benefits significantly from polynomial multiplication optimizations and present a detailed memory profile. In constrained environments, Dilithium's signature generation is notably slower than verification, but its overall computational cost is lower than schemes like SPHINCS+. Despite larger key sizes and signatures compared to ECC, its performance and security make it suitable for firmware signing and software attestation in edge devices.

Castiglione et al. [162] achieved optimized implementation of Dilithium-5 on ESP32 microcontroller and also developed a prototype for quantum resistant Heart rate and Blood oxygen level measurement device. Two separate ESP32 modules have been used, one for data capture and another dedicatedly for Signature generation and verification resulting faster computation time. Further they integrated the Dilithium-5 implementation to blockchain network for transaction processing. Implementation on ESP32 and computational and storage analysis confirmed the suitability of the scheme to lightweight devices.

c: FALCON

FALCON (Fast Fourier Lattice-based Compact Signatures over NTRU) is a signature scheme based on the NTRU lattice and the Short Integer Solution (SIS) problem. It achieves compact signatures through Fast Fourier sampling and Gaussian lattice sampling. Unlike Dilithium, FALCON uses floating-point arithmetic and requires careful constant-time implementation of Gaussian sampling for security against side-channel attacks. It offers two parameter sets: FALCON-512 and FALCON-1024, targeting NIST security levels 1 and 5, respectively.

Zia et al. [160] show that FALCON has the smallest signature size among all NIST PQC finalists, but is the most complex to implement securely, especially on embedded processors without hardware floating-point units. The authors evaluate fixed-point emulation of floating-point operations and report performance bottlenecks in Falcon's signature generation phase. Despite these challenges, Falcon is attractive in bandwidth-constrained applications (e.g., authentication over low-speed links) but requires carefully engineered platforms.

Nguyen et al. [163] present a high-speed hardware accelerator architecture for CRYSTAL-Kyber and CRYSTAL-

Dilithium, optimizing polynomial multiplication via Number Theoretic Transform (NTT) on resource-constrained platforms. The proposed design introduces a novel DSP architecture that reduces the critical path delay of modular multiplication and eliminates the need for separate pre- and post-processing steps. Furthermore, unified pipelined architectures for both Radix-2 and Radix-4 NTT/INTT transformations are developed, achieving full throughput utilization and reducing redundant memory access. Implemented on an Artix-7 FPGA, their accelerator achieves a speed-up of up to $34.16\times$ for Kyber and $9.6\times$ for Dilithium, with a corresponding 44%–96% improvement in area-time-product. These results highlight the suitability of their design for secure, low-latency IoT and embedded applications resistant to quantum and side-channel attacks.

CRYSTALS-Kyber has seen significant deployment efforts on embedded platforms, particularly microcontrollers such as the ARM Cortex-M4 [43]. In research and implementation efforts like the pqm4 project, Kyber has been successfully optimized in C and assembly to run efficiently on Cortex-M4 processors (e.g., STM32F4 series), which are widely used in IoT and embedded systems [31], [164], [165]. These optimizations focus on speeding up the number-theoretic transform (NTT), a central component of Kyber's computation, achieving practical performance within the constraints of limited CPU and memory. In parallel, hardware-accelerated implementations of Kyber have been demonstrated on FPGAs (e.g., Xilinx Artix-7 or Zynq) and System-on-Chip platforms. For CRYSTALS-Dilithium, the story is quite similar. However, it involves a more complex signing process than Kyber, it benefits from shared hardware components like NTT cores and hash accelerators [31], [165]. FPGA-based platforms such as KiD (Kyber and Dilithium) reuse common logic blocks, significantly reducing hardware area while supporting both key exchange and signature operations. On microcontrollers, Dilithium has been implemented with careful optimization to fit within the performance and memory limits of devices like the ARM Cortex-M4. As with Kyber, it is also supported via the `liboqs` (Open Quantum Safe) project, allowing integration into TLS libraries and OS-level services, making it deployable on embedded Linux platforms that power modern IoT gateways and constrained devices [43].

Falcon, due to its reliance on fast Fourier transforms and complex sampling algorithms, has historically been harder to implement efficiently on constrained devices. However, a variant called Falcon-M has been proposed for microcontrollers such as Cortex-M4 and RISC-V platforms. This version simplifies trapdoor sampling to reduce computational overhead, trading off some theoretical security margin for practical deployability. Moreover, Falcon has been successfully accelerated on desktop and GPU platforms, with tools like GOLF demonstrating ultra-high throughput on NVIDIA RTX series GPUs. This makes Falcon particularly suitable for high-speed digital signature generation and verification in backend servers or cloud-based TLS infrastructures,

while Falcon-M makes initial steps toward embedded deployment.

Across all three schemes, researchers have proposed and tested implementations on platforms like Contiki OS, a lightweight operating system tailored for IoT devices [164]. While full integration is still in progress for most PQC algorithms, preliminary work shows that with code optimizations and hardware acceleration, post-quantum security is achievable even in constrained environments [43], [165]. As standardization progresses and microcontroller vendors begin offering native PQC support, these schemes are likely to be widely embedded in future secure IoT deployments.

To make it more intuitive, Table 10 shows how these post-quantum cryptographic (PQC) algorithms integrate within hardware and software stacks.

2) N^{TH} TRUNCATED RING POLYNOMIAL UNIT (NTRU)

NTRU is a ring-based public key crypto-system widely used ever since its introduction in 1998 [166]. NTRU is based on the SVM and not on factorization. It is experimentally found to be twice as fast as the RSA algorithm and is quantum-effective.

Many researchers have worked extensively to apply lattice-based cryptographic methods suitable to IoT applications. We discuss a few recent contributions in the IoT context. Several contributions prefer to use Ring-LWE rather than LWE errors, as the key size in R-LWE is significantly smaller compared to LWE techniques that guarantee the same level of security. These proposed algorithms are operated on suitable and supporting hardware that complements their functionalities.

Ebrahimi et al. [167] deeply analyzed the vulnerabilities of Bin-RLWE [168] and proposed necessary improvements to them. The improved scheme is resilient to fault and timing attacks; however, it takes longer for encryption and decryption. The authors handled every coefficient directly in 2's complement notation to improve speed by avoiding the need for any additional reduction operations following modular arithmetic.

Khalid et al. [169] designed a lightweight co-processor suitable for IoT applications using Ring-LWE. This scheme employs binomial sampling instead of a Gaussian distribution and, as a result, is more hardware-friendly. It facilitates various approximation levels for the given dimension of the lattice. It is an apt alternative to ECC in terms of cost and efficiency.

Qian et al. [170] devised an encryption scheme based on cipher-text policy attribute-based encryption. The scheme has excellent resilience to quantum attacks with an optimization scheme. The protocol employs small key sizes, making it adaptable for IoT applications while allowing the re-usability of secret keys.

Xiang et al. [171] showed that decryption processes that use a discrete Gaussian distribution might give wrong results. To overcome this issue, the authors have proposed to use the

TABLE 10. Deployment of PQC algorithms across platforms.

Function	ARM Cortex-M4 (e.g., STM32)	RISC-V / SoC / FPGA	Desktop / GPU / Cloud
Key Encapsulation (Kyber)	C + assembly optimized via NTT, runs in $\sim 10^5$ – 10^6 cycles on STM32 microcontrollers	NTT modules in FPGA (e.g., Artix-7, Zynq) with shared cores for Kyber + Dilithium	AES/NEON/AVX2-accelerated implementations; supported in liboqs; integrated into TLS libraries
Signature (Dilithium)	Microcontroller and FPGA-friendly; uses shared NTT cores for area-efficient hardware	Shared NTT hardware; software fallback via liboqs in embedded Linux environments	Highly optimized for AVX2/SSE; used in TLS and kernel-level crypto via liboqs
Signature (Falcon)	Falcon-M variant written in C fits Cortex-M4 class devices; lightweight trapdoor variant	Possible via FPGA soft-core implementations, but not widely deployed yet	GPU-parallelized (e.g., GOLF on NVIDIA RTX); also CPU-optimized using liboqs

Ring-BinLWE scheme, which is based on 2's complement ring.

Xu et al. [172] presented a lightweight Ring-EXPLWE encryption scheme that replaced the previously used binary distribution technique with exponential distribution. The scheme offers a higher security level with minimal hardware overhead.

Homomorphic encryption (HE) is a subset of ring-learning from errors that allows users to perform certain additive or multiplicative operations on the encrypted text itself without the need to decrypt the cipher text into plain text. Qui et al. [173] proposed a FHE algorithm to preserve the privacy of the IoT edge nodes and cloud servers. The designed system incorporates identity-based encryption (IBE) to ensure the privacy of various participating IoT nodes.

Agarwal et al. [174] introduced a data extraction from an image recognition technique to be employed in Smart grids using Partial HE. The decentralized system architecture described in this research combines time stamping and Boneh – Lynn – Shacham (BLS) signatures with IoT-based image analysis and a partly homomorphic cryptosystem that is resistant to replay attack and False Data Injection (FDI) attack. As a result, it efficiently preserves privacy while maintaining data integrity and source authentication.

Li et al. [175] proposed a HE-based technique to ensure privacy and security in resource-constrained applications. In the IIoT, a simple privacy-preserving method is presented that divides computing expenses into a fixed and a dynamic portion, with only the dynamic part being updated during data processing. This scheme cuts down on computation overheads at a single node and is as a result, apt for lightweight processing at edge nodes.

Agus et al. [176] proposed NTRU encryption in data communication of IoT applications. The scheme claims to be very resource-efficient and has the least communication overhead.

Jaime et al. [177] carried out an experiment to test the performance of NTRU encapsulation in communication within the Contiki-NG operating system. The experiment reveals that NTRU performs best in modern micro-controllers; however, NTRU increases the spatial overhead in terms of RAM usage and may be efficient in extreme edge nodes.

Gudipalli et al. [178] introduced NTRU encryption, which can be used in data communication. As the Wireless Fidelity (Wi-Fi) Protocols no longer guarantee a secure network between the AP and the STA, [178] proposed an NTRU-based authentication and encryption technique for Wi-Fi security. The recommended encryption algorithm has the advantage of being faster than RSA and ECC at key generation, decryption, and encryption. In addition to the advantage of getting a rapid rate with fewer computing resources, the NTRU is devoid of problems with factoring or discontinuous logarithms.

Singamaneni et al. [179] came up with a hybrid post quantum scheme for encryption and access control using NTRUEncrypt-SIDH and KP-ABE. The scheme is targeted for 6G Industry 5.0 applications and supports DDoS attack resistance measures such as rate limiting, traffic filtering and adaptive routing. With encryption, decryption and key generation times of 1.8 s/GB, 1.667 s/GB and 2.857 s/GB and overall time of 11.7 ms the scheme is suitable for resource constrained devices.

An interesting work regarding implementation of NTTRU on ARMv8 was proposed in [180]. They managed to achieve speedup of 2.85x on keygeneration, 2.36x on encapsulation and 3.27 times on decapsulation using NEON engine of ARMv8. Further they integrated the proposed implementation of NTTRU KEM with TLS1.3 and evaluated the TLS handshake performance against other lattice based KEM. For backward compatibility they explored a hybrid implementation of the scheme with ECC and found it to be the fastest among other comparable schemes.

B. CODE-BASED CRYPTOGRAPHY

McEllice [148] proposed a public key encryption scheme based on decoding linear codes, a well-known NP-hard problem. The encryption scheme is able to accept random parameters. Code-based schemes are based on matrices rather than factorization and, as a consequence, are computationally much faster than traditional cryptosystems.

Three algorithms, such as Key Generation (KeyGen), Encapsulation (Encaps), and Decapsulation (Decaps), combine to produce KEMs, a public-key cryptographic system. KEMs are specialized Public Key Encryption (PKEs) schemes frequently constructed utilizing transformations

like the Fujisaki Okamoto Transform. Unlike the Diffie-Hellman Key Exchange, they are not intended to do any data encryption but rather to provide a shared secret that is randomly generated between communication partners in cryptographic protocols like Transport Layer Security (TLS). Then, using quick symmetric cryptographic techniques like Advanced Encryption Standard (AES), this shared secret is used to create a secret session key for decrypting and encrypting application data. Schoffel et al. [181] stated that code-based approaches require larger memory overheads but involve more mathematically complex problems. As a result, they promise better security than lattice-based algorithms. The authors in [181] devised a hardware and software-coupled mechanism for code-based Hamming Quasi Cyclic KEM (HQC) to strengthen the security of the edge nodes. The results demonstrate that the implementation has good and efficient security.

C. MULTIVARIATE-BASED CRYPTOGRAPHY

Multivariate polynomials are based on NP hard and NP complete problems defined over a finite field [33]. There are specific situations in which these polynomials can be defined across an extension field. Multivariate quadratics are what we refer to when the polynomials have a degree of two [182]. Multivariate schemes have the obvious benefit of providing post-quantum security.

Akleyek et al. [183] stated that digital signatures can be used to mitigate attacks pertaining to unauthorized access. The authors presented a multivariate signature based on multivariate polynomials that is immune to attacks from quantum computers and has effective key and signature sizes. Due to the proposed scheme's excellent parallelism support, edge node authentication may be deployed effectively at high throughput. The proposed system can create 113 signatures per second and verify 120 signatures per second when implemented on a GPU device, which is quicker than a serial implementation on a CPU.

Debnath et al. [184] introduced an identity-based signature scheme (IBS) to address the content poisoning attacks in the IoT network, namely Mul-IBS relying on multivariate public key cryptography (MPKC). Their primary contribution is to provide content security in named data networking (NDN). IoT systems work well with the Mul-IBS since it is a very lightweight signature scheme. It reduces the risk of content tampering by providing solid security assurances along with limited and affordable computing resources. Mul-IBS's lightweighness makes it possible to keep caches free of outdated information. An IoT-based NDN system with a strong and durable cryptographic foundation may be created using the proposed Mul-IBS design.

D. ISOGENY KEY EXCHANGE

Isogeny-based key exchange is a promising approach to post-quantum cryptography that relies on the hardness of computing isogenies between supersingular elliptic curves.

Specifically, given two supersingular elliptic curves E_1 and E_2 defined over a finite field $\mathbb{F}_q$, it is computationally difficult to find an isogeny $\phi : E_1 \rightarrow E_2$. This difficulty forms the basis of the Supersingular Isogeny Diffie-Hellman (SIDH) protocol, a quantum-secure analog of the traditional Diffie-Hellman key exchange [185].

In SIDH, each party generates a secret isogeny from a shared starting curve and exchanges partial public information, specifically images of torsion basis points under their secret isogeny. The combination of these public values allows both parties to independently compute a shared secret. A notable strength of SIDH is its compact key size compared to other post-quantum schemes [185].

However, SIDH's security was compromised in July 2022 when Castryck and Decru [186] introduced a key recovery attack. Their method exploited the auxiliary torsion point images shared during the protocol, in combination with the known endomorphism ring of the supersingular curve, and applied Kani's reducibility criterion to reconstruct the secret isogeny.

To address this vulnerability, recent research emphasizes using curves with unknown endomorphism rings, as seen in the SECUER approach [185]. By instantiating SIDH over such random curves, the endomorphism structure becomes hidden, thereby mitigating attacks like the Castryck-Decru method.

Moreover, the extended SIDH variant (eSIDH) introduces algorithmic improvements to enhance computational performance. These include optimized prime forms that allow for parallelism, implementations on Montgomery and Edwards curve models, and support for multicore computation. As shown in [185], eSIDH achieves better efficiency while still adhering to the principles of isogeny-based security.

While SIDH and its variant SIKE are no longer considered secure in their original forms, the broader field of isogeny-based cryptography remains active. Future directions involve refining protocols to obscure structural information and designing key exchange mechanisms resilient against quantum and classical cryptanalysis.

Table 11 summarizes the advances on post-quantum Schemes.

VIII. DISCUSSION ON RESEARCH GAP

A. CONTRAST BETWEEN TRADITIONAL PKC AND POST-QUANTUM PKC

Before the advent of quantum computers, traditional public key cryptography (PKC) was more prevalent. Shor's algorithm revolutionized cryptography by efficiently solving the factorization problem, posing significant challenges to traditional PKC systems. There are attempts to mitigate these challenges by increasing key sizes, but this approach introduces delays in key generation and imposes greater computational overhead. However, in resource-constrained IoT applications, enlarging key sizes is impractical. Hence, the need for quantum-resistant cryptographic systems has become imperative.

TABLE 11. Recent advances in IoT Security based on Post-Quantum Security.

Reference	Year	Application	Type	Time	Evaluation metrics	Attack mitigated	Lightweight	Security Scheme
Ebrahimi et al. [167]	2020	-	-	153 ms	Clock cycles, Computational complexity and Time	Quantum Fault and Timing attacks	Yes	Bin-RLWE
Agus et al. [176]	2020	-	-	6 ms	Computational overhead	Quantum Masquerading attacks	Yes	NTRU
Qui et al. [173]	2020	-	-	-	Computation cost	Quantum Fault attacks	Yes	IBE-HE
Khalid et al. [169]	2021	-	-	-	Area, Power, Energy consumption	Quantum attacks	Yes	AxRLWE
Agarwal et al. [174]	2021	Smart grid	Image	60 ms	Computation, Communication cost	Data disclosure	No	Partial - HE
Li et al. [175]	2021	IIoT	Sensor reading	151 ms	Computation, Communication cost	Data disclosure, Collusion	Yes	HE
Qian et al. [170]	2021	-	-	-	Security, Computation, Communication cost	Data disclosure, Impersonation	Yes	LWE
Xiang et al. [171]	2022	-	-	-	Clock Cycles, Decryption approximation, Communication cost	Eavesdropping, Data-theft, Leakage	Yes	Binary Ring-LWE
Xu et al. [172]	2022	-	-	53.4 ms	Clock Cycles, Computation cost	Eavesdropping, Data-theft, Leakage	Yes	Ring-EXPLWE
Jaime et al. [177]	2022	-	-	-	Clock Cycles, Computation, Storage Cost	Hardware attacks -	No	NTRU
Schoffel et al. [181]	2022	-	-	-	Clock Cycles, Computation, Storage Cost	Hardware attacks -	Yes	Code-based HQC
Gudipalli et al. [178]	2023	-	-	-	Encryption Time, Security Strength	Eavesdropping, Impersonation	Yes	NTRU Encrypt
Akleyek et al. [183]	2020	-	-	-	Clock Cycles, Security Strength, Throughput	Impersonation	Yes	Multivariate-Signature
Zia et al. [160]	2022	Embedded IoT	Signature, KEM	Kyber: 313 μ s, Dilithium: 3.2ms (sign), Falcon: 1.9ms (sign)	Runtime, Energy, Clock cycles	Quantum, Side-channel	Yes	Kyber, Dilithium, Falcon
Debnath et al. [184]	2023	NDN	-	-	Security Strength	Repudiation, Impersonation	Yes	Multivariate - IBS
Kim et al. [159]	2024	Embedded IoT	Signature, KEM	Kyber: 370 μ s, Dilithium: 1.4ms (sign)	Runtime, NTT speedup, Keygen/Verify time	Timing attacks, side-channel	Yes	Kyber, Dilithium
Nguyen et al. [163]	2024	Embedded IoT	Signature, KEM	Kyber: 149 CCs, Dilithium: 128 CCs	Latency, Area-Time Product, Frequency	Timing, Side-channel	Yes	Kyber, Dilithium
Castiglione et al. [162]	2025	IoT Devices	Signature	Signature: 438 ms, Verification: 72 ms	Computation cost storage cost	Fault analysis, Side channel	Yes	Dilithium-5
Singamaneni et al. [179]	2025	6G Industry 5.0	Encryption, access control	11.7 ms	Computational cost	Distributed DoS	Yes	NTRUEncrypt-SIDH, KP-ABE
Saeed. [156]	2025	IoT Devices	Anomaly detection, integrity, encryption	-	Response effectiveness score	Time delayed injection, gradual data poisoning, packet mimicry	Yes	SHA-256, LWE
Zhang et al. [180]	2025	Embedded IoT	KEM	Keygen: 62185 CCs, Encaps: 57555 CCs Decaps: 52111 CCs	Computation cost, packet loss	Quantum attacks	Yes	NTTRU, TLS1.3
Torre et al. [161]	2025	Wireless communication	PKE, KEM	WKyber512 v1: 140 CCs Wkyber512 v2: 140.8 CCs	Error probability, computational cost	Dual, Primal	No	Kyber IND-CCA IND-CPA

In contrast, post-quantum cryptography is not reliant on factorization or discrete logarithm problems and is resistant to quantum attacks. It addresses NP-hard problems, offering resistance against quantum algorithms. While post-quantum methods generally exhibit shorter encryption and decryption times, they require more storage and longer key generation times due to the complexity of their underlying mathematical problems.

It's worth noting that post-quantum cryptography, while considered quantum-safe, is not impervious to all attacks. For instance, lattice-based attacks can compromise post-quantum systems if larger prime values are chosen. Additionally, post-quantum solutions tend to be intricate and challenging

to understand due to the complexity of the mathematical problems they rely on. The 2022 attack on the SIDH protocol [186] demonstrated a full key recovery using a novel application of Kani's reducibility criterion. This attack exploited structural weaknesses in SIDH, specifically the public exposure of torsion point images and the use of starting elliptic curves with known endomorphism rings. It broke all known SIDH parameter sets—including SIKEp434, a NIST Level 1 candidate—in under 10 minutes on a single core. As a result, SIDH and its instantiation SIKE were withdrawn from post-quantum standardization efforts. This vulnerability reshapes the narrative around "quantum safety" by highlighting that post-quantum cryptographic systems,

despite being resistant to quantum algorithms like Shor's, may still be vulnerable to powerful classical attacks.

The field of cryptography has seen advancements in hybrid cryptography, integrating both traditional and post-quantum methods to enhance security and robustness. For example, Mustafa et al. [187] introduced a lattice-based version of RSA, replacing integer factorization. Their scheme consists of vector factorization, vector mapping, and finding the shortest vector within an n -dimensional lattice. Simulation results in AVISPA demonstrate that their scheme maintains security even in the presence of intruders.

In response to the cryptographic challenges posed by quantum computing, the U.S. National Institute of Standards and Technology (NIST) has selected CRYSTALS-Kyber and CRYSTALS-Dilithium as primary algorithms for post-quantum public-key encryption and digital signatures, respectively. Kyber is an IND-CCA2-secure key encapsulation mechanism (KEM) based on the hardness of the Module Learning With Errors (Module-LWE) problem, a structured lattice-based assumption offering both strong security guarantees and high performance. It supports three parameter sets—Kyber-512, Kyber-768, and Kyber-1024—corresponding approximately to classical AES-128, AES-192, and AES-256 security levels. Due to its efficiency and relatively small key sizes, Kyber is being adopted in hybrid modes alongside classical schemes such as elliptic-curve Diffie-Hellman, with Kyber-768 recommended for conservative 128-bit quantum-resistant security. On the digital signature front, NIST has endorsed CRYSTALS-Dilithium, also based on structured lattices, as the primary algorithm due to its favorable balance between signature size, computational cost, and security. While FALCON is also recommended for use cases demanding smaller signatures, and SPHINCS+ is retained as a fallback due to its hash-based security model, Dilithium is expected to serve as the default in most deployments. Both Kyber and Dilithium have seen early adoption by industry players including Cloudflare, Amazon Web Services, and IBM, indicating strong ecosystem support and readiness for integration in post-quantum cryptographic infrastructures. There are also challenges like vulnerability of Kyber and Dilithium to side channel attacks such as timing and power attacks, but research is going on to provide potential solutions using constant time algorithm, hardware security modules, masking, blinding and fault tolerant designs.

B. INTEGRATION CHALLENGES OF 5G AND PQC

The integration of PQC into 5G networks must account for operational characteristics that differ from previous cellular generations. While PQC offers asymptotic security benefits against quantum-capable adversaries, its performance implications are non-trivial in the context of 5G's latency, bandwidth, and device constraints.

The 5G standard supports three primary service categories: ultra-reliable low-latency communications (URLLC),

enhanced mobile broadband (eMBB), and massive machine-type communications (mMTC) [188]. URLLC scenarios are particularly sensitive to handshake latency, as target end-to-end delays may be below one millisecond, leaving minimal budget for additional cryptographic processing or transmission overhead. mMTC devices typically operate under strict memory, processing, and energy limitations, which may be incompatible with large PQC key sizes or computationally expensive primitives [189]. eMBB scenarios, while less latency-sensitive, must sustain high throughput and frequent connection establishments, where larger authentication exchanges may increase control-plane load.

The nature of the application's data and its alert latency requirements further influence PQC suitability. Low-bandwidth, delay-tolerant workloads, such as periodic environmental sensor readings in an mMTC slice, can accommodate schemes with larger public keys, ciphertexts, or slower operations, as the relative overhead to the payload is small and transmission frequency is low. In such cases, implementation simplicity or maximising long-term security may take precedence over minimising size. Conversely, high-bandwidth applications, such as continuous medical imaging in an eMBB slice, are more sensitive to per-packet size increases, since PQC overhead can contribute to substantial aggregate bandwidth expansion and potential fragmentation at the network layer. Latency-sensitive alerting systems, such as collision warnings in autonomous driving or critical health monitoring alarms, impose strict upper bounds on end-to-end delay, often in the sub-10 millisecond range. For such use cases, only schemes with very low encapsulation, decapsulation, signing, and verification latency are acceptable, even if this constrains the choice of algorithms or parameter sets [188].

Mobility-related operations such as handovers require repeated authentication and key establishment. The increased size of PQC handshake messages can exacerbate signalling congestion. Many lattice-based and code-based PQC schemes produce public keys, signatures, or ciphertexts in the kilobyte range, which in some cases exceed the path maximum transmission unit, leading to fragmentation [188]. On wireless links, fragmentation increases retransmission probability, which negatively impacts reliability and latency. Architectural features such as network slicing and multi-access edge computing influence deployment choices: terminating encryption at the edge can reduce cryptographic latency, but this may compromise end-to-end quantum resistance unless the edge is treated as part of the trust boundary.

A deployment offering complete post-quantum protection at all layers maximises security against a future quantum adversary but introduces measurable costs in terms of bandwidth, latency, and energy. Hybrid key exchange, for example combining a classical elliptic curve Diffie-Hellman with a PQC key encapsulation mechanism, provides security against both classical and quantum threats but increases handshake message size and computational load [189]. Selecting higher

NIST security categories increases resistance margins but further inflates parameter sizes, which may exceed the capabilities of certain mMTC devices.

Potential strategies to align PQC adoption with 5G operational constraints include selecting schemes with favourable size–performance characteristics for constrained links, employing session resumption and early data modes to reduce the frequency of full handshakes while implementing replay-mitigation measures, utilising hardware acceleration for PQC operations in both user equipment and base station hardware, applying certificate compression and handshake optimisation to limit additional control-plane signalling, and differentiating PQC profiles by network slice, applying stronger security configurations to slices carrying critical URLLC or control-plane traffic and more size-optimised configurations for mMTC slices.

Incorporating PQC into 5G networks therefore requires a context-aware deployment strategy that considers both network-level constraints and application-specific performance requirements, balancing long-term cryptographic resilience with the stringent operational demands of next-generation mobile systems. Table 12 summaries the application profiles in 5G and implications for PQC scheme selection.

C. OPEN RESEARCH ISSUES

The pace of growth of research is indeed commendable; however, there are still areas that require attention:

- 1) *Comprehensive Threat Mitigation*: Current solutions address specific threats like DoS or impersonation but often fail to protect data confidentiality comprehensively. No solution effectively covers all potential threats in IoT systems.
- 2) *Practicality for IoT Networks*: Many proposed solutions involve complex key generation and encryption processes that can slow down real-time systems. In IoMT applications, such delays might lead to critical failures, underscoring the need for efficient, low-latency methods. The complexity of PQC schemes poses a major challenge. However, hybrid schemes such as those implemented in Open Quantum Safe (liboqs) provide an incremental deployment strategy. These allow gradual migration from classical to post-quantum schemes in TLS protocols, reducing disruption and maintaining practical performance in real-time systems like IoMT.
- 3) *Lightweight Implementation*: While some solutions offer lightweight versions for low-resource systems, they often require specific hardware or parallel implementations, which can be impractical. As 5G enables distributed machine learning and inference at the edge, lightweight cryptographic primitives must be compatible with edge AI workloads. PQC schemes like Falcon-M or Dilithium-2 can be deployed on micro-controllers (e.g., ARM Cortex-M4) using optimized C and assembly implementations. On FPGA-based edge platforms such as Xilinx Zynq, AI accelerators can share logic blocks (e.g., Number Theoretic Transform units) with Kyber and Dilithium to reduce area overhead and power consumption. These cryptographic services should be tightly integrated with lightweight real-time operating systems such as Contiki-NG or Zephyr, enabling secure communication among edge devices with minimal latency.
- 4) *Complexity vs. Efficiency Trade-offs*: Post-quantum solutions involve vector convolution operations, necessitating careful selection of coefficients to balance security and performance.
- 5) *Node Tampering and Fault Tolerance*: Current solutions lack adequate focus on node tampering and fault tolerance, which are crucial for ensuring the reliability of edge nodes. AI-enhanced trust scoring can offer fault detection capabilities by flagging compromised nodes based on deviations in behavior. Additionally, federated learning systems with secure aggregation can maintain resilience by enabling redundancy in model updates and minimizing the trust placed on any single node.
- 6) *Hybrid Cryptography Integration*: Integrating pre and post-quantum security methods could enhance IoT security. Further it makes the scheme backward compatible with pre quantum cryptography. However, research on effective hybrid approaches is limited, and future work should aim to combine these methods effectively. Practically, hybrid key exchange and signature mechanisms can be implemented using platforms such as Open Quantum Safe (liboqs), which integrates with OpenSSL to provide hybrid TLS handshakes. For resource-constrained devices, embedded TLS libraries (e.g., ARM mbedTLS) can be patched to support PQC primitives. These implementations are particularly relevant for secure boot protocols, over-the-air firmware updates, and mutual authentication between edge devices and gateways in 5G networks.
- 7) *Energy-Efficient 5G IoT*: Research should focus on developing low-power communication protocols, energy harvesting methods, and resource allocation strategies to extend device battery life and reduce overall energy consumption.
- 8) *AI-driven Optimization in 5G IoT*: Exploring AI and machine learning applications for optimizing 5G-enabled IoT systems could enhance device provisioning, network resource management, and service orchestration. Research could include advanced AI algorithms for dynamic network optimization, predictive analytics for maintenance, and autonomous decision-making. With the increasing adoption of federated and collaborative learning in IoT ecosystems, it is essential to integrate cryptographic safeguards that preserve data confidentiality and model integrity. Post-quantum secure multiparty computation (MPC) protocols can enable devices to jointly compute AI models without

TABLE 12. Application profiles in 5G and implications for PQC scheme selection (Kyber for KEM, Dilithium for signatures).

Application profile	Data bandwidth	Latency requirement	5G slice	Implications for PQC scheme selection
Periodic environmental sensing	Low (bytes–kilobytes per message)	Delay-tolerant (seconds–minutes)	mMTC	Kyber-512 and Dilithium-II suitable; larger parameter sets acceptable if long-term security is prioritised over efficiency; handshake size not critical.
Industrial sensor networks with periodic control	Low–moderate	Moderate (≤ 100 ms)	mMTC / eMBB (private network)	Kyber-512 or Kyber-768 with Dilithium-II; balance efficiency and moderate key size to avoid excess signalling load.
Continuous medical imaging	High (tens–hundreds of Mbps)	Moderate (≤ 100 ms)	eMBB	Kyber-768 with Dilithium-III recommended for stronger security; prioritise compact signatures to minimise per-packet expansion; hardware acceleration beneficial.
Autonomous vehicle collision warnings	Low–moderate	Ultra-low (< 10 ms)	URLLC	Kyber-512 with Dilithium-II to minimise computational delay; pre-shared keys or session resumption advisable to avoid full handshakes in critical alerts.
Real-time health monitoring alarms	Low	Low (< 50 ms)	URLLC	Kyber-512 with Dilithium-II preferred; hybrid KEMs only if processing and size overheads fit within latency budget.

revealing raw inputs. Signatures based on Dilithium or Falcon can be used to validate model updates in federated learning pipelines, ensuring provenance and protecting against model poisoning or rollback attacks. One of the major research challenge is the availability of the 5G specific dataset for training the models. To reduce the computational load researchers are exploring pretrained models and LLMs for anomaly detection, malware detection and automated patching of vulnerabilities. The concept of explainable AI for more transparent AI decisions is also gaining much attention in the AI based crypto solutions like IDS. AI based traffic analysis and optimization is also being used in efficient implementation of 5G building blocks like network slicing and beamforming. Privacy preservation using GAN based data augmentation, use of DRL to deal with security issues involving high dimensional states and implementation of zero touch security management are some of the latest use of AI in 5g based IoT system [41], [42].

IX. CONCLUSION

The Internet of Things has had a great impact on revolutionizing our lifestyle. It connects objects embedded with sensors to transmit useful information or percepts about an environmental parameter. It helps in remote monitoring and surveillance operations in high-altitude and deep oceans. This extensive survey has aimed to offer a perceptive summary of the current status of research in the field of IoT security, especially considering the upcoming era of quantum computing. We begin with the meaning of IoT, extend our discussion to its applications and security vulnerabilities, and explore the broad classes of solutions to address the issues. Furthermore, we investigated the growing significance of 5G technology in enhancing the functionalities of IoT applications, leading to era of ultra-low latency and high bandwidth connectivity. The combination

of 5G with IoT offers potential for enhanced data analysis, preventive maintenance, and efficient use of resources in various fields. We summarize our discussion with a contrast between pre and post quantum public-key cryptosystems and point out a few open issues in this context. Advent of quantum computing has challenged the long standing use of public key cryptographic algorithms, thus beginning the era of post quantum cryptography (PQC). But resource constrained nature of IoT devices has brought significant challenges to the adoption of PQC in 5G based IoT systems. However already concerted efforts are being made for adaptation of PQC for lightweight IoT devices and significant progress may be seen in the near future.

CONFLICT OF INTEREST

The authors do not have any relevant financial or non-financial interests to disclose.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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